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(54) GENETICALLY MODIFIED YEAST AND FERMENTATION PROCESSES FOR THE PRODUCTION OF ARABITOL

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(57) **ABSTRACT**

Disclosed herein are genetically engineered yeast cells capable of producing arabitol. The engineered yeast cell may comprise an exogenous polynucleotide sequence encoding an arabitol 2-dehydrogeanse (ARD2DH) enzyme comprising a sequence at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 95%, at least 98%, at least 99%, or 100% identical to at least one of SEQ ID NOs: 1, 2, 3, 9, or 11.

Specification includes a Sequence Listing.

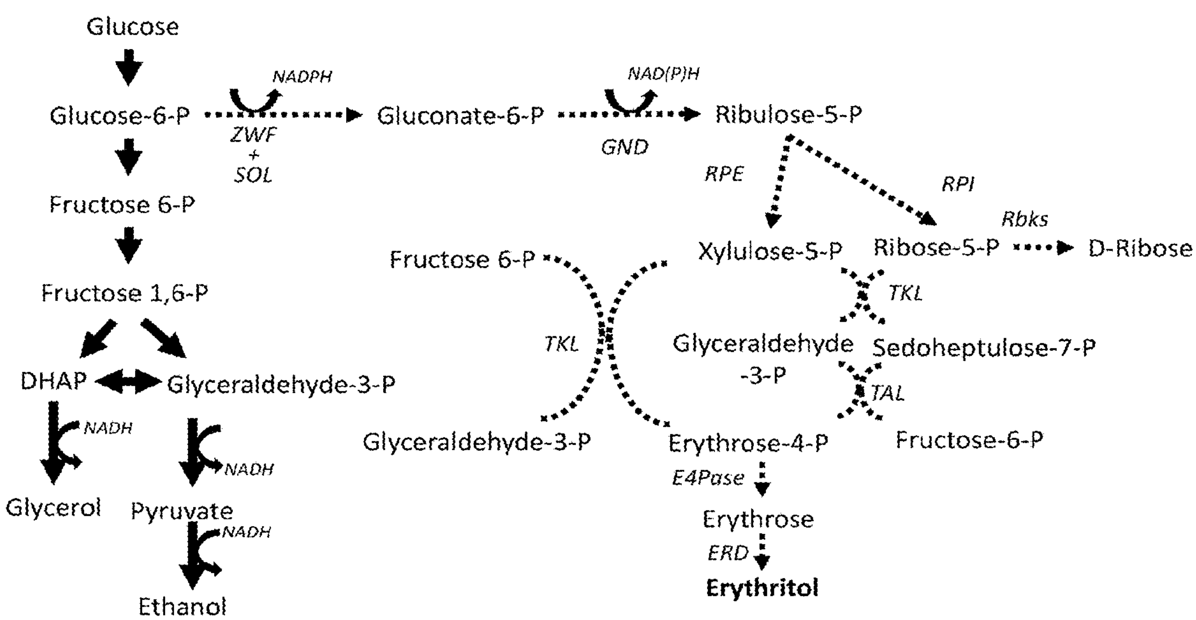


FIG. 1

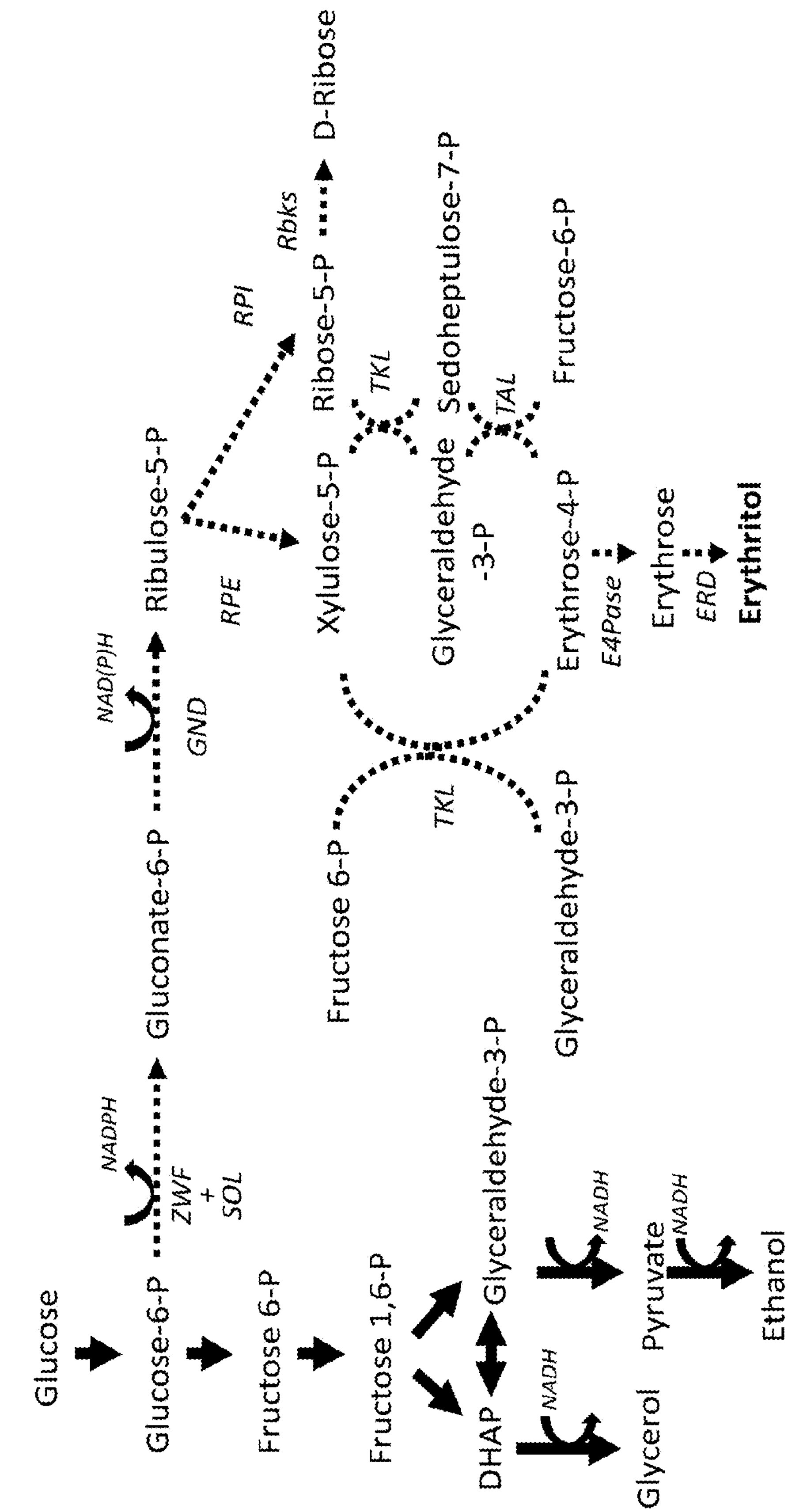

Glucose
Glucose-6-P
Fructose 6-P
Fructose 1,6-P
DHAP
Glyceraldehyde-3-P
NADH
Glycerol
Pyruvate
Ethanol
NADPH
ZWF
+
SOL
Gluconate-6-P
NAD(P)H
GND
Ribulose-5-P
RPE
RPI
Xylulose-5-P
Ribose-5-P
Rbks
D-Ribose
TKL
Sedoheptulose-7-P
Glyceraldehyde
-3-P
TAL
Fructose-6-P
Erythrose-4-P
Fructose 6-P
Glyceraldehyde-3-P
E4Pase
Erythrose
ERD
Erythritol

FIG. 2

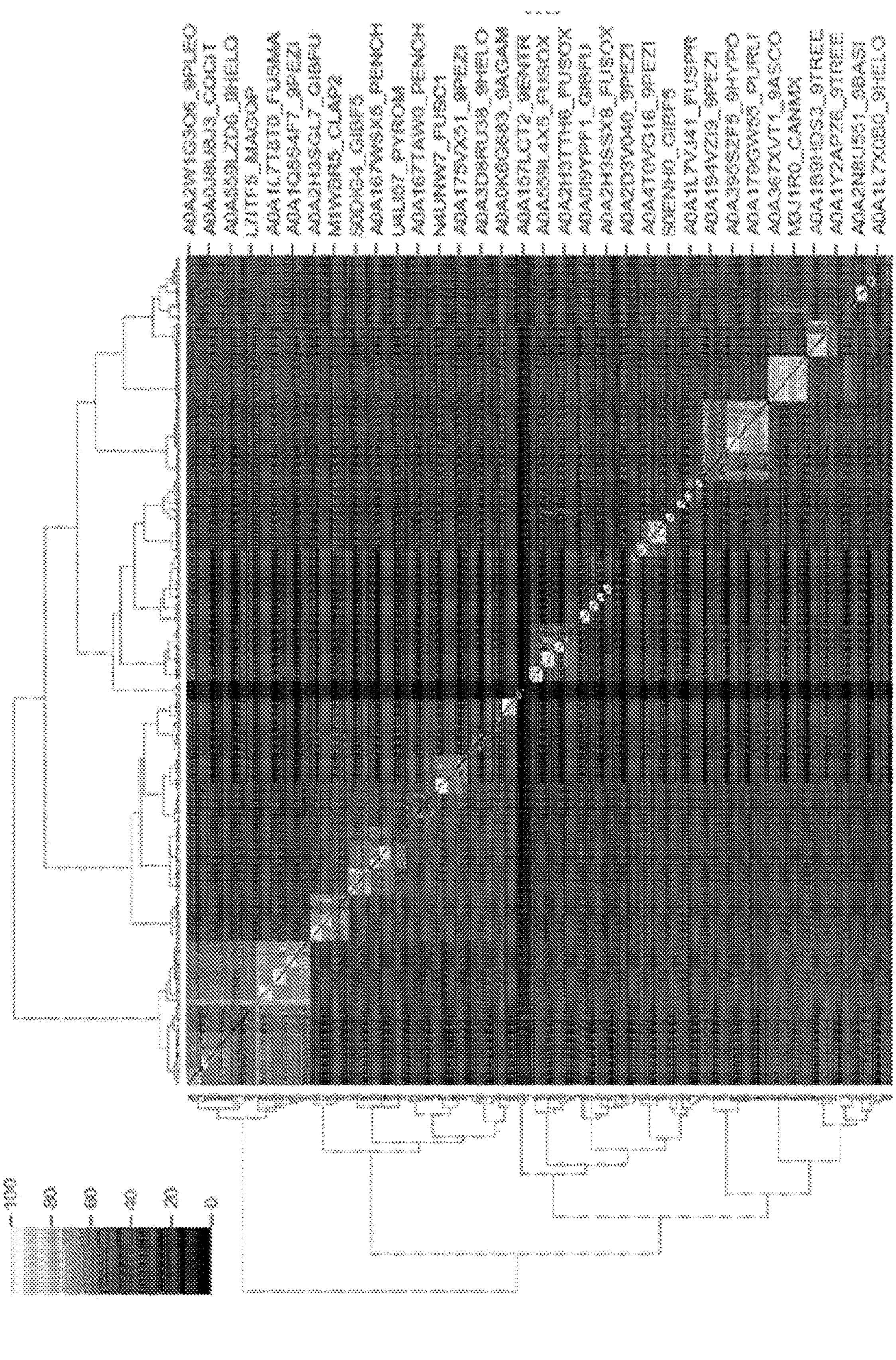
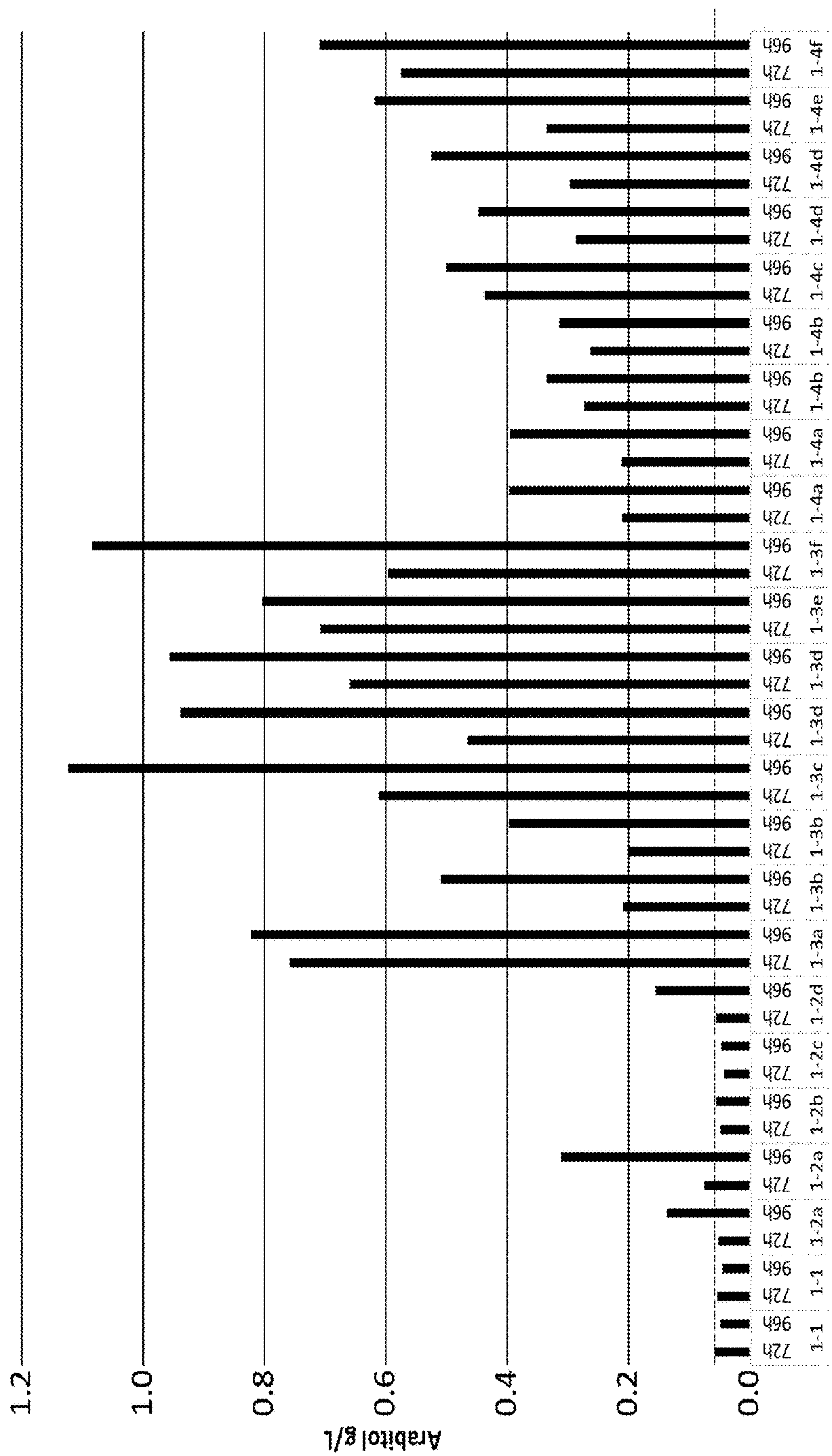


FIG. 3



Arabitol g/L
0.0
0.2
0.4
0.6
0.8
1.0
1.2
72h
96h
1-1
1-2a
1-2b
1-2c
1-2d
1-3a
1-3b
1-3c
1-3d
1-3e
1-3f
1-4a
1-4b
1-4c
1-4d
1-4e
1-4f

FIG. 4

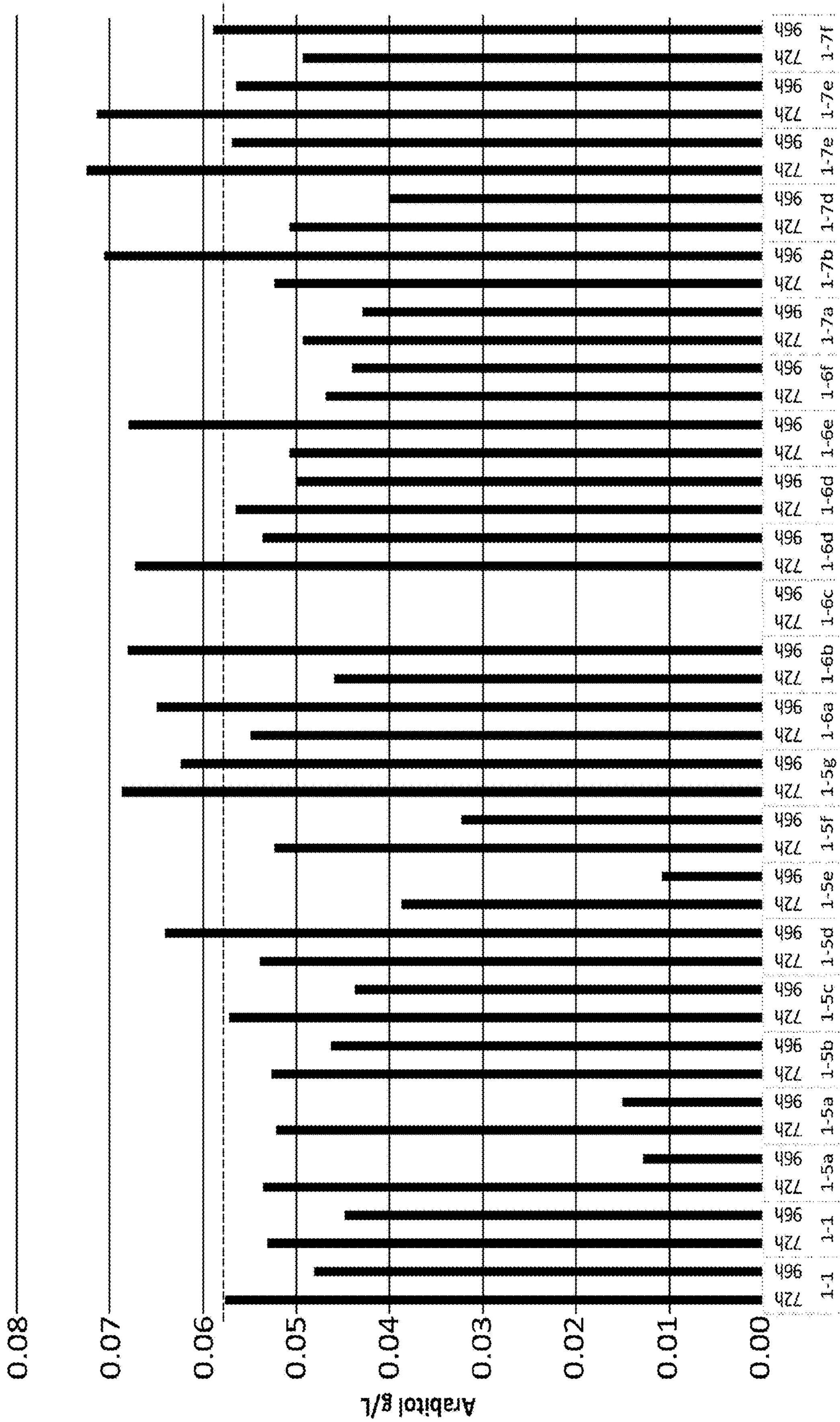
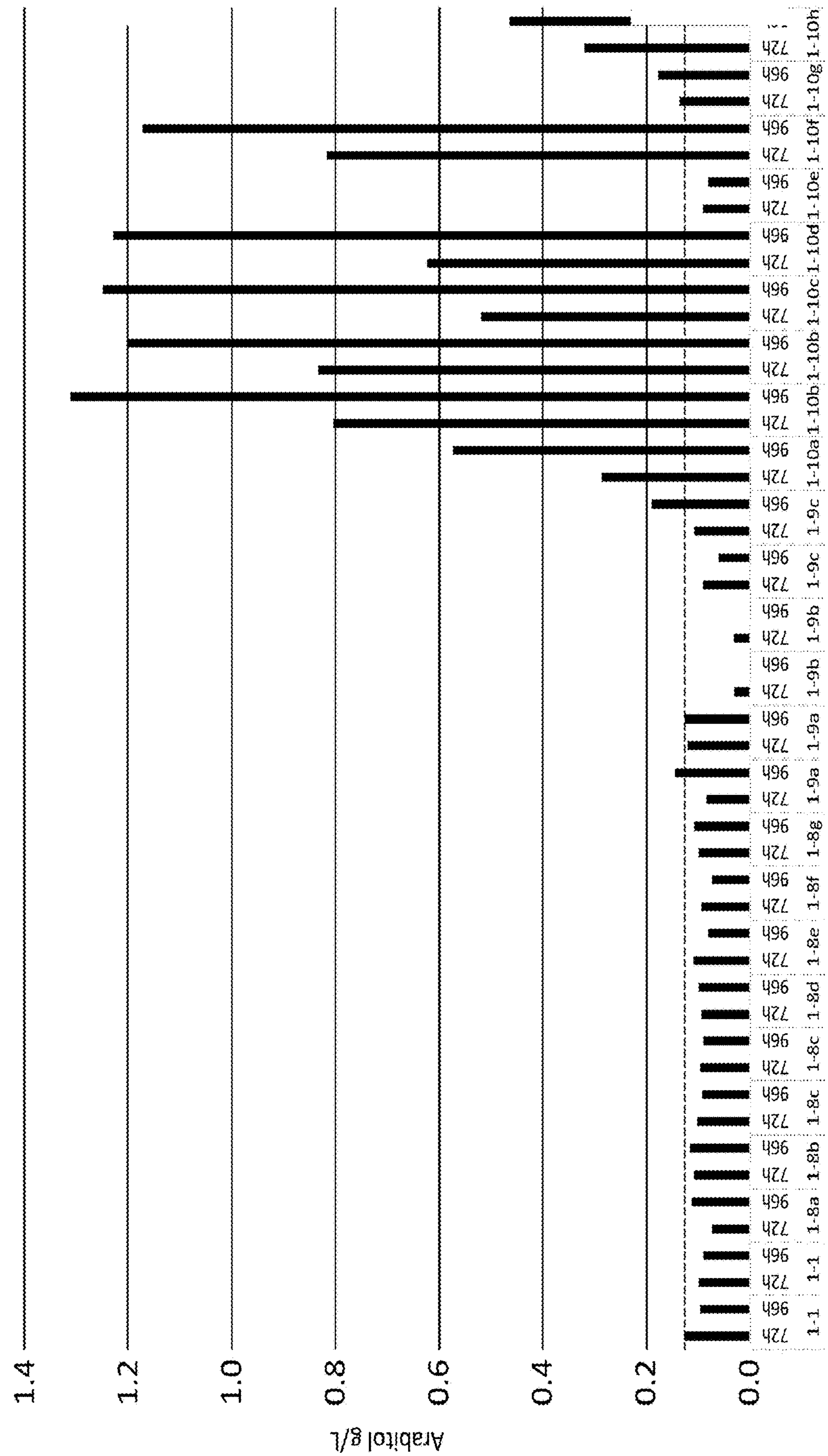
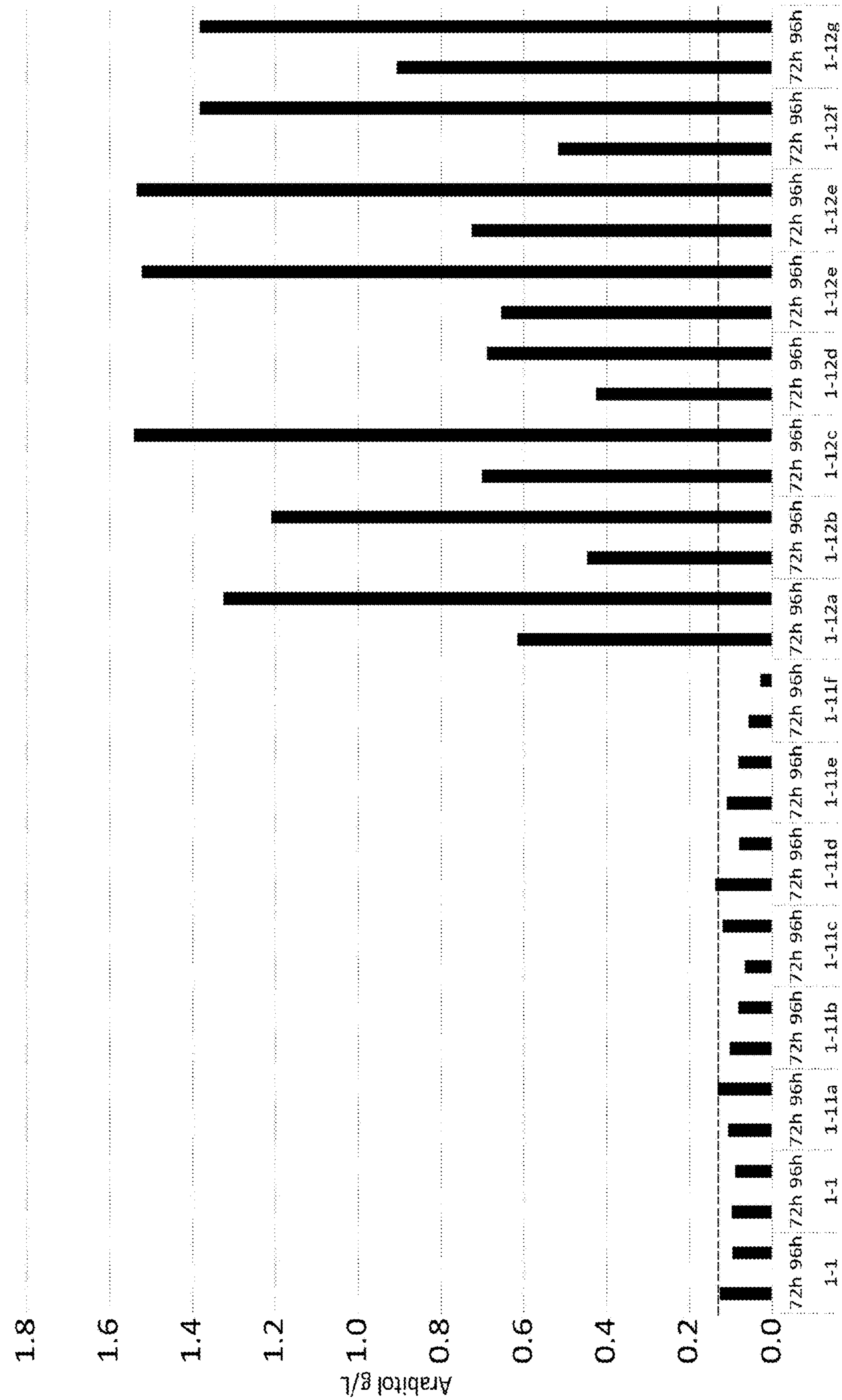

Arabitol g/L
0.00
0.01
0.02
0.03
0.04
0.05
0.06
0.07
0.08
72h 96h 1-1
72h 96h 1-1
72h 96h 1-5a
72h 96h 1-5a
72h 96h 1-5b
72h 96h 1-5c
72h 96h 1-5d
72h 96h 1-5e
72h 96h 1-5f
72h 96h 1-5g
72h 96h 1-6a
72h 96h 1-6b
72h 96h 1-6c
72h 96h 1-6d
72h 96h 1-6d
72h 96h 1-6e
72h 96h 1-6f
72h 96h 1-7a
72h 96h 1-7b
72h 96h 1-7d
72h 96h 1-7e
72h 96h 1-7e
72h 96h 1-7f

FIG. 5



Arabitol g/L
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1.4
72h
96h
1-1
1-8a
1-8b
1-8c
1-8d
1-8e
1-8f
1-8g
1-9a
1-9b
1-9c
1-10a
1-10b
1-10c
1-10d
1-10e
1-10f
1-10g
1-10h

FIG. 6


Arabitol g/L
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1.4
1.6
1.8
72h 96h
1-1
72h 96h
1-1
72h 96h
1-11a
72h 96h
1-11b
72h 96h
1-11c
72h 96h
1-11d
72h 96h
1-11e
72h 96h
1-11f
72h 96h
1-12a
72h 96h
1-12b
72h 96h
1-12c
72h 96h
1-12d
72h 96h
1-12e
72h 96h
1-12e
72h 96h
1-12f
72h 96h
1-12g

GENETICALLY MODIFIED YEAST AND FERMENTATION PROCESSES FOR THE PRODUCTION OF ARABITOL

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application claims the benefit of U.S. Provisional Application No. 63/364,359, filed May 9, 2022, which is incorporated by reference herein in its entirety.

REFERENCE TO A SEQUENCE LISTING SUBMITTED VIA PATENT CENTER

[0002] The content of the Sequence Listing XML file of the sequence listing named "PT-1351-WO-PCT.xml" which is 150,554 bytes in size created on May 4, 2023 and electronically submitted via Patent Center herewith the application is incorporated by reference in its entirety.

BACKGROUND

[0003] Xylitol is a low-calorie sweetener used as a food additive and sugar substitute. Commonly used in drug, dietary supplement, confectionary, and toothpaste compositions, xylitol has also been associated with anticariogenic properties when used in chewing gums. Traditional methods of xylitol production, including chemically catalyzed hydrogenation of xylose hydrolyzed from biomass extracted xylan, are both monetarily and environmentally costly. These methods require high temperatures and pressures, large amounts of water, and metal catalysts that must be mined. In contrast, fermentation processes have been used commercially at large scale to produce other organic molecules, such as ethanol, citric acid, lactic acid, and the like, and may offer a cost effective and sustainable alternative to traditional xylitol processing methods.

[0004] In the development of microorganism-based fermentation strategies for the production of xylitol, production of metabolic pathway intermediates and alternative fermentation products are important considerations. For example, metabolic pathways active in the production of xylitol may have overlap with the metabolic pathways for the production of arabitol, erythritol, ribitol, and the like. The intermediates and products have their own uses and markets that make their fermentation commercially relevant. Accordingly, provided herein are genetically modified yeast and fermentation methods for the production of arabitol.

SUMMARY

[0005] The present disclosure provides a genetically engineered yeast cell capable of producing arabitol, the engineered yeast cell comprising an exogenous polynucleotide sequence encoding an arabitol 2-dehydrogeanse (ARD2DH) enzyme comprising a sequence at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 95%, at least 98%, at least 99%, or 100% identical to at least one of SEQ ID NOs:1, 2, 3, 9, or 11. The yeast cell may be an osmotolerant yeast cell. The yeast cell may be a cell of the subphylum Ustilaginomycotina. The yeast cell may be selected form the group consisting of *Trichosporonoides megachiliensis*, Trychosporonoides *oedocephalis*, Trychosporonoides *nigrescens, Pseudozyma tsukubaensis, Trigonopsis variabilis, Moniliella*, Ustilaginomycetes, *Trichosporon, Yarrowia lipolytica, Penicillium, Torula, Pichia, Candida, Candida magnoliae*, and *Aureobasidium*. The yeast cell may be a yeast cell of the genus.

[0006] The disclosure also provides a genetically engineered *Moniliella* cell capable of producing arabitol, the engineered *Moniliella* cell comprising an exogenous polynucleotide sequence encoding an arabitol 2-dehydrogeanse (ARD2DH) enzyme comprising a sequence at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 95%, at least 98%, at least 99%, or 100% identical to at least one of SEQ ID NOs:1, 2, 3, 9, or 11.

[0007] The ARD2DH enzyme may have a sequence at least 85% identical to SEQ ID NO:2, 3, 9, and/or 11. The ARD2DH enzyme may have a sequence at least 90% identical to SEQ ID NO: 2, 3, 9, and/or 11.

[0008] The engineered cell described herein may be a *Moniliella pollinis* cell. The yeast cell described herein may be capable of producing arabitol at a titer of at least 0.2, 0.5, 0.75, 1.0, 1.5, or 2.0 g/L when used in a fermentation process in the presence of dextrose at 35° C. for 96 hours. Erythritol production by the engineered cell described herein may be reduced relative to erythritol production in an equivalent yeast cell lacking the exogenous polynucleotide sequence.

[0009] The exogenous polynucleotide sequence may be operably linked to a heterologous or artificial promoter. The promoter may be a constitutive promoter. The promoter may be selected from the group consisting of pyruvate kinase 1 promoter (PYK1p; SEQ ID NO:49), 6-phosphogluconate dehydrogenase promoter (6PGDp; SEQ ID NO:40), glyceraldehyde-3-phosphate dehydrogenase promoter (TDH3p; SEQ ID NO:42), translational elongation factor 1 promoter (TEFp; SEQ ID NO:43), modified TEFp (SEQ ID NO:41), phosphoglucomutase 1 promoter (PGM1p; SEQ ID NO:44), 3-phosphoglycerate kinase promoter (PGK1p; SEQ ID NO:45), enolase promoter (ENO1p; SEQ ID NO:46), asparagine synthetase promoter (ASNSp; SEQ ID NO:47), 50S ribosomal protein L1 promoter (RPLAp; SEQ ID NO:48), and RPL16B (SEQ ID NO:50).

[0010] The disclosure also provides a method for producing arabitol using the engineered cells described herein, the method comprising contacting a substrate comprising dextrose with an engineered cell described herein, wherein fermentation of the substrate by the engineered yeast produces arabitol. The disclosure also provides a method for producing arabitol, the method comprising contacting a substrate comprising dextrose with an engineered yeast cell comprising an exogenous polynucleotide sequence encoding an arabitol 2-dehydrogeanse (ARD2DH) enzyme comprising a sequence at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 95%, at least 98%, at least 99%, or 100% identical to at least one of SEQ ID NOs:1, 2, 3, 9, or 11, wherein fermentation of the substrate by the engineered yeast produces arabitol. The fermentation temperature may be at or between 25° C. to 45° C., 30° C. to 40° C., or 32° C. to 37° C. The volumetric oxygen uptake rate (OUR) may be between 0.5 to 40, 1 to 35, 2 to 30, 3 to 25, 4 to 20, or 5 to 15 mmol $O_2/(L \cdot h)$. Erythritol production may be reduced relative to an equivalent fermentation run with an equivalent yeast cell lacking the exogenous polynucleotide sequence. Erythritol production may be less than 60, 50, 40, or less than 30 g/L when the fermentation is run at 35° C. for 96 hours. Arabitol production may be at least 0.2, 0.5, 0.75, 1.0, 1.5, or 2.0 g/L when the fermentation is run at 35° C. for 96 hours. Glycerol production may be reduced relative to an equivalent fermentation run with an equivalent yeast cell lacking the exogenous polynucleotide sequence. Ethanol production may be reduced relative to an equivalent fermentation run with an equivalent yeast cell lacking the exogenous polynucleotide sequence.

BRIEF DESCRIPTION OF THE FIGURES

[0011] This patent or application contains at least one drawing executed in color. Copies of this patent or patent application publication with color drawings will be provided by the Office upon request and the payment of the necessary fee.

[0012] The drawings illustrate generally, by way of example, but not by way of limitation, various aspects discussed herein.

[0013] FIG. **1** shows the predicted native pentose phosphate pathway (dotted lines and arrows) and the native glycolysis pathways (solid lines and arrows) in *Moniliella pollinis*.

[0014] FIG. **2** shows diversity in the arabitol 2-dehydrogenase (ARD2DH) sequence space.

[0015] FIG. **3** shows arabitol concentrations (g/L) at 72 hours and 96 hours of shake flask fermentation as outlined in Example 3. The dotted line shows the level of arabitol production in the parent strain 1-1.

[0016] FIG. **4** shows arabitol concentrations (g/L) at 72 hours and 96 hours of shake flask fermentation as outlined in Example 3. The dotted line shows the level of arabitol production in the parent strain 1-1.

[0017] FIG. **5** shows arabitol concentrations (g/L) at 72 hours and 96 hours of shake flask fermentation as outlined in Example 4. The dotted line shows the level of arabitol production in the parent strain 1-1.

[0018] FIG. **6** shows arabitol concentrations (g/L) at 72 hours and 96 hours of shake flask fermentation as outlined in Example 4. The dotted line shows the level of arabitol production in the parent strain 1-1.

DETAILED DESCRIPTION

[0019] Reference will now be made in detail to certain aspects of the disclosed subject matter, examples of which are illustrated in part in the accompanying drawings. While the disclosed subject matter will be described in conjunction with the enumerated claims, it will be understood that the exemplified subject matter is not intended to limit the claims to the disclosed subject matter.

[0020] In this document, the terms "a," "an," or "the" are used to include one or more than one unless the context clearly dictates otherwise. The term "or" is used to refer to a nonexclusive "or" unless otherwise indicated. All publications, patents, and patent documents referred to in this document are incorporated by reference herein in their entirety, as though individually incorporated by reference. In the event of inconsistent usages between this document and those documents so incorporated by reference, the usage in the incorporated reference should be considered supplementary to that of this document; for irreconcilable inconsistencies, the usage in this document controls.

[0021] Values expressed in a range format should be interpreted in a flexible manner to include not only the numerical values explicitly recited as the limits of the range, but also to include all the individual numerical values or sub-ranges encompassed within that range as if each numerical value and sub-range were explicitly recited. For example, a range of "about 0.1% to about 5%" or "about 0.1% to 5%" should be interpreted to include not just about 0.1% to about 5%, but also the individual values (e.g., 1%, 2%, 3%, and 4%) and the sub-ranges (e.g., 0.1% to 0.5%, 1.1% to 2.2%, 3.3% to 4.4%) within the indicated range. The statement "about X to Y" has the same meaning as "about X to about Y," unless indicated otherwise. Likewise, the statement "about X, Y, or about Z" has the same meaning as "about X, about Y, or about Z," unless indicated otherwise.

[0022] Unless expressly stated, ppm (parts per million), percentage, and ratios are on a by weight basis. Percentage on a by weight basis is also referred to as wt % or % (wt) below.

[0023] This disclosure relates to various recombinant cells engineered to produce arabitol. In general, the recombinant cells described herein have an active pentose phosphate pathway and are characterized by expression of an exogenous arabitol 2-dehydrogenase (ARD2DH) enzyme. The disclosure further provides fermentation methods for the production of arabitol from dextrose using the genetically engineered cells described herein.

[0024] In general, recombinant cells described herein are yeast cells. As used herein, "yeast" refers to eukaryotic single celled microorganisms classified as members of the fungus kingdom. Yeast are unicellular organisms which evolved from multicellular ancestors with some species retaining multicellular characteristics such as forming strings of connected budding cells known as pseudo hyphae or false hyphae. Yeast cells may also be referred to in the art as yeast-like cells, and as used herein "yeast cell" encompasses both yeast and yeast-like cells. Suitable yeast and yeast-like host cells for modification may include, but are not limited to, *Saccharomyces cerevisiae*, Komagataella sp., *Kluyveromyces* (e.g., *Kluyveromyces lactis, Kluveromyces marxianus*), *Yarrowia lipolytica, Issatchenkia orientalis, Pichia galeiformis, Pichia* sp. YB-4149 (NRRL designation), *Pichia pastoris, Candida* (e.g., *Candida magnoliae, Candida ethanolica*), *Pichia deserticola, Pichia membranifadens, Pichia fermentans, Aspergillus, Trichoderma*, Myceliphthora *thermophila, Moniliella* (e.g., *Moniliella pollinis*), *Pfaffia, Yamadazyma, Hansenula, Pichia* kudriavzevvi, *Trichosporonoides* (e.g., *Trichosporonoides megachiliensis*, Trychosporonoides *oedocephalis*, Trychosporonoides *nigrescens*), *Pseudozyma tsukubaensis, Trigonopsis variabilis, Penicillium*, and *Torula*. An ordinarily skilled artisan would understand the requirements for selection of a suitable yeast cell, and recombinant yeast cells of the present disclosure are not limited to those expressly recited herein. Methods for genetic engineering of yeast cells are known and described in the art and a skilled artisan would understand the methods necessary to transform and engineer a suitable yeast cell.

[0025] A suitable yeast cell may be a cell of the phylum Basidiomycota and the subphylum Ustilaginomycotina. Suitable yeast of the subphylum Ustilaginomycotina include, but are not limited to, *Ustilago* (e.g., *U. cynodontis, U. maydis, U. sphaerogena, U. cordal, U. scitaminea, U. coicis, U. syntherismae, U. esculenta, U. neglecta, U. crus-galli, Ustilago avenae*), *Sporisorium* (e.g., *Sporisorium exsertum*), *Moniliella* (e.g., *M. pollinis, M. tomentosa, M. acetoabutans, M. fonsecae, M. madida, M. megachiliensis, M. ocedocephalis, M. nigrescens*), and *Pseudozyma* (e.g., *Pseudozyma tsukubaensis*), and *Trichosporonoides* (e.g., *Trichosporonoides megachiliensis*, Trychosporonoides *oedocephalis*, Trychosporonoides *nigrescens*). Yeast of the subphylum Ustilaginomycotina have been known and described in the art as potential production organisms for valuable chemicals such as itaconate, malate, succinate, mannitol, and erythritol and other valuable biotechnological applications. See, for example, Geiser et al. (Prospecting the biodiversity of the fungal family Ustilaginacceae for the production of value-added chemicals," Fungal Biol Biotechnol, 2014, 1:2), Feldbrugge et al., ("The biotechnological use and potential of plant pathogenic smut fungi," Appl Microbiol Biotechnol, 2013, 97(8):3253-65), Guevarra et al., ("Accumulation of itaconic, 2-hydroxyparaconic, itatartaric, and malic acids by strains of the genus *Ustilago*, Agric. Biol. Chem., 1990, 54(9), 2353-2358), and Moon et al., ("Biotechnological production of erythritol and its applications," Appl Microbiol Biotechnol, 2010, 86:1017-1025).

[0026] A suitable yeast cell will have an active pentose phosphate pathway that produces ribulose-5-phosphate. As used herein "active pentose phosphate pathway" refers to expression of one or more functional enzymes which, together, convert glucose-6-phosphate, $NADP^+$ or NAD+, and water to NADPH or NADH, CO2, and ribulose-5-phosphate. Continuing in a non-oxidative phase, the pathway may also produce other pentose (i.e., 5-carbon) sugars. For example, the pentose phosphate pathway may produce ribulose-5-phosphate, ribose-5-phosphate, xylulose-5-phosphate, fructose 6-phosphate, combinations thereof, and the like, depending on the enzymatic activities present. The active pentose phosphate pathway may be native to the yeast cell or it may be introduced into the yeast cell by genetic engineering.

[0027] The yeast cell may be an osmotolerant yeast cell. As used herein, "osmotolerant" refers to a yeast capable of growth and reproduction under conditions of high osmolarity, such as at least 10% (w/v), at least 20% (w/v), at least 30% (w/v), at least 40% (w/v), at least 50% (w/v), or at least 60% (w/v) glucose and/or at least 6% (w/v), at least 10% (w/v), at least 12% (w/v), at least 13% (w/v), at least 15% (w/v) sodium chloride. Species and strains of osmotolerant yeast are known and described in the art, including many species of yeast used in industrial fermentation processes. Likewise, methods for assaying yeast osmotolerance are known and described in the art. See, for example, Tiwari, S., et al., ("Nectar yeast community of tropical flowering plants and assessment of their osmotolerance and xylitol-producing potential," Current Microbiology, 2022, 79:28).

[0028] The recombinant yeast cell may be a recombinant *Moniliella* cell, for example, a *Moniliella pollinis* cell. FIG. **1** shows the predicted native pentose phosphate and glycolysis pathways in *Moniliella pollinis*. *Moniliella* has previously been used in the fermentation production of erythritol and methods for genetically modifying and fermenting *Moniliella* are known and described in the art. See, for example, Li et al. ("Methods for genetic transformation of filamentous fungi," 2017, Microb Cell Fact, 16:168).

[0029] Various plasmids and methods for transformation of *Moniliella* are also described in the Examples below. For example, *Moniliella* may be transformed using a bipartite polynucleotide sequence(s) in which, following recombination, the exogenous polynucleotide of interest is integrated at the specified locus and the selection marker is expressible within the cell. Suitable selection markers are known and used in the art. The selectable marker may include, but is not limited to, amdS (for example broken into a 3' portion, SEQ ID NO:63, and a 5' portion, SEQ ID NO:64), G418 resistance gene (for example broken into a 3' portion, SEQ ID NO:69, and a 5' portion, SEQ ID NO:70), zeocin resistance gene (for example broken into a 3' portion, SEQ ID NO:65, and a 5' portion, SEQ ID NO:66), nourseothricin N-acetyl transferase (NAT) (for example broken into a 3' portion, SEQ ID NO:67, and a 5' portion, SEQ ID NO:68), and invertase gene (SUC2) (for example a 3' portion of SEQ ID NO:71 and a 5' portion of SEQ ID NO:72).

[0030] The recombinant cells described herein include one or more exogenous polynucleotide sequences encoding one or more exogenous polypeptides that, when expressed improve the fermentation of glucose to ribitol by the recombinant cells.

[0031] The terms "glucose" and "dextrose" are used interchangeably herein and refer to D-glucose except where expressly indicated otherwise.

[0032] As used herein, "exogenous" refers to genetic material or an expression product thereof that originates from outside of the host organism. For example, the exogenous genetic material or expression product thereof can be a modified form of genetic material native to the host organism, it can be derived from another organism, it can be a modified form of a component derived from another organism, or it can be a synthetically derived component. For example, a *K. lactis* invertase gene is exogenous when introduced into *S. cerevisiae.*

[0033] As used herein, "native" refers to genetic material or an expression product thereof that is found, apart from individual-to-individual mutations which do not affect function or expression, within the genome of wild-type cells of the host cell. For the purposes of this application, the *Moniliella pollinis* cell "*Moniliella tomentosa* var *pollinis* TCV364" described in U.S. Pat. No. 6,440,712, which is incorporated herein by reference in its entirety, and deposited under the Budapest Treaty at BCCM/MUCL (Belgian Coordinated Collections of Micro-organisms/Mycothéque de l'Université Catholique de Louvain by *Eridania* Béghin Say, Vilvoorde R&D Centre, Havenstraat 84, B-1800 Vilvoorde) on Mar. 28, 1997 under number MUCL40385, is considered the wild-type *Moniliella pollinis* cell.

[0034] As used herein, the terms "polypeptide" and "peptide" are used interchangeably and refer to the collective primary, secondary, tertiary, and quaternary amino acid sequences and structure necessary to give the recited macromolecule its function and properties. As used herein, "enzyme" or "biosynthetic pathway enzyme" refer to a protein that catalyzes a chemical reaction. The recitation of any particular enzyme, either independently or as part of a biosynthetic pathway is understood to include the co-factors, co-enzymes, and metals necessary for the enzyme to properly function. A summary of the amino acids and their three and one letter symbols as understood in the art is presented in Table 1. The amino acid name, three letter symbol, and one letter symbol are used interchangeably herein.

TABLE 1

Amino Acid three and one letter symbols

Amino Acid	Three letter symbol	One letter symbol
Alanine	Ala	A
Arginine	Arg	R

TABLE 1-continued

Amino Acid three and one letter symbols		
Amino Acid	Three letter symbol	One letter symbol
Asparagine	Asn	N
Aspartic acid	Asp	D
Cysteine	Cys	C
Glutamic acid	Glu	E
Glutamine	Gln	Q
Glycine	Gly	G
Histidine	His	H
Isoleucine	Ile	I
Leucine	Leu	L
Lysine	Lys	K
Methionine	Met	M
Phenylalanine	Phe	F
Proline	Pro	P
Serine	Ser	S
Threonine	Thr	T
Tryptophan	Trp	W
Tyrosine	Tyr	Y
Valine	Val	V

[0035] Variants or sequences having substantial identity or homology with the polypeptides described herein can be utilized in the practice of the disclosed recombinant cells, compositions, and methods. Such sequences can be referred to as variants or modified sequences. That is, a polypeptide sequence can be modified yet still retain the ability to exhibit the desired activity. Generally, the variant or modified sequence may include greater than about 45%, 45%, 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, or 95% sequence identity with the wild-type, naturally occurring polypeptide sequence, or with a variant polypeptide as described herein.

[0036] As used herein, the phrases "% sequence identity," "% identity," and "percent identity," are used interchangeably and refer to the percentage of residue matches between at least two amino acid sequences or at least two nucleic acid sequences aligned using a standardized algorithm. Methods of amino acid and nucleic acid sequence alignment are well-known. Sequence alignment and generation of sequence identity include global alignments and local alignments which are carried out using computational approaches. An alignment can be performed using BLAST (National Center for Biological Information (NCBI) Basic Local Alignment Search Tool) version 2.2.31 software with default parameters. Amino acid % sequence identity between amino acid sequences can be determined using standard protein BLAST with the following default parameters: Max target sequences: 100; Short queries: Automatically adjust parameters for short input sequences; Expect threshold: 10; Word size: 6; Max matches in a query range: 0; Matrix: BLOSUM62; Gap Costs: (Existence: 11, Extension: 1); Compositional adjustments: Conditional compositional score matrix adjustment; Filter: none selected; Mask: none selected. Nucleic acid % sequence identity between nucleic acid sequences can be determined using standard nucleotide BLAST with the following default parameters: Max target sequences: 100; Short queries: Automatically adjust parameters for short input sequences; Expect threshold: 10; Word size: 28; Max matches in a query range: 0; Match/Mismatch Scores: 1, −2; Gap costs: Linear; Filter: Low complexity regions; Mask: Mask for lookup table only. A sequence having an identity score of XX % (for example, 80%) with regard to a reference sequence using the NCBI BLAST version 2.2.31 algorithm with default parameters is considered to be at least XX % identical or, equivalently, have XX % sequence identity to the reference sequence.

[0037] Polypeptide or polynucleotide sequence identity may be measured over the length of an entire defined polypeptide sequence, for example, as defined by a particular SEQ ID number, or may be measured over a shorter length, for example, over the length of a fragment taken from a larger, defined polypeptide sequence, for instance, a fragment of at least 15, at least 20, at least 30, at least 40, at least 50, at least 70 or at least 150 contiguous residues. Such lengths are exemplary only, and it is understood that any fragment length supported by the sequences shown herein, in the tables, figures or Sequence Listing, may be used to describe a length over which percentage identity may be measured.

[0038] The polypeptides disclosed herein may include "variant" polypeptides, "mutants," and "derivatives thereof." As used herein the term "wild-type" is a term of the art understood by skilled persons and means the typical form of a polypeptide as it occurs in nature as distinguished from variant or mutant forms. As used herein, a "variant," "mutant," or "derivative" refers to a polypeptide molecule having an amino acid sequence that differs from a reference protein or polypeptide molecule. A variant or mutant may have one or more insertions, deletions, or substitutions of an amino acid residue relative to a reference molecule.

[0039] The amino acid sequences of the polypeptide variants, mutants, derivatives, or fragments as contemplated herein may include conservative amino acid substitutions relative to a reference amino acid sequence. For example, a variant, mutant, derivative, or fragment polypeptide may include conservative amino acid substitutions relative to a reference molecule. "Conservative amino acid substitutions" are those substitutions that are a substitution of an amino acid for a different amino acid where the substitution is predicted to interfere least with the properties of the reference polypeptide. In other words, conservative amino acid substitutions substantially conserve the structure and the function of the reference polypeptide. Conservative amino acid substitutions generally maintain (a) the structure of the polypeptide backbone in the area of the substitution, for example, as a beta sheet or alpha helical conformation, (b) the charge and/or hydrophobicity of the molecule at the site of the substitution, and/or (c) the bulk of the side chain.

[0040] As used herein, terms "polynucleotide," "polynucleotide sequence," and "nucleic acid sequence," and "nucleic acid," are used interchangeably and refer to a sequence of nucleotides or any fragment thereof. These phrases also refer to DNA or RNA of natural or synthetic origin, which may be single-stranded or double-stranded and may represent the sense or the antisense strand. The DNA polynucleotides may be a cDNA (e.g., coding DNA) or a genomic DNA sequence (e.g., including both introns and exons).

[0041] A polynucleotide is said to encode a polypeptide if, in its native state or when manipulated by methods known to those skilled in the art, it can be transcribed and/or translated to produce the polypeptide or a fragment thereof. The anti-sense strand of such a polynucleotide is also said to encode the sequence.

[0042] Those of skill in the art understand the degeneracy of the genetic code and that a variety of polynucleotides can encode the same polypeptide. In some aspects, the polynucleotides (i.e., polynucleotides encoding an ARD2DH polypeptide) may be codon-optimized for expression in a particular cell including, without limitation, a plant cell, bacterial cell, fungal cell, or animal cell. While polypeptides encoded by polynucleotide sequences found in various species are disclosed herein any polynucleotide sequences may be used which encodes a desired form of the polypeptides described herein. Thus, non-naturally occurring sequences may be used. These may be desirable, for example, to enhance expression in heterologous expression systems of polypeptides or proteins. Computer programs for generating degenerate coding sequences are available and can be used for this purpose. Pencil, paper, the genetic code, and a human hand can also be used to generate degenerate coding sequences.

[0043] The recombinant cells described herein may include deletions or disruptions in one or more native genes. The phase "deletion or disruption" refers to the status of a native gene in the recombinant cell that has either a completely eliminated coding region (deletion) or a modification of the gene, its promoter, or its terminator (such as by a deletion, insertion, or mutation) so that the gene no longer produces an active expression product, produces severely reduced quantities of the expression product (e.g., at least a 75% reduction or at least a 90% reduction) or produces an expression product with severely reduced activity (e.g., at least 75% reduced or at least 90% reduced). The deletion or disruption can be achieved by genetic engineering methods, forced evolution, mutagenesis, RNA interference (RNAi), and/or selection and screening. The native gene to be deleted or disrupted may be replaced with an exogenous nucleic acid of interest for the expression of an exogenous gene product (e.g., polypeptide, enzyme, and the like).

[0044] The recombinant cells described herein may include one or more genetic modifications in which an exogenous nucleic acid is integrated into the genome of the host cell. One of skill in the art know how to select suitable loci in a yeast genome for integration of the exogenous nucleic acid. Suitable integration loci may include, but are not limited to, the PDC1, GPD1, CYB2A, CYB2B, g4240, YMR226, MDHB, ATO2, Adh9091, Adh1202, ADE2, ADH2556, GAL6, MDH1, SCW11, ER1, ER3, pyrF, TRP3, gpdIIA, and gpdIIB loci. For example, in a *M. pollinis* host cells, suitable interaction loci may include, but are not limited to, the ER1 locus (defined as the locus flanked by SEQ ID NO:36 and SEQ ID NO:37), the ER3 locus (defined as the locus flanked by SEQ ID NO:24 and SEQ ID NO:25), the PDC1 locus (defined as the locus flanked by SEQ ID NO:26 and SEQ ID NO:27), the pyrF locus (defined as the locus flanked by SEQ ID NO:28 and SEQ ID NO:29), the TRP3 locus (defined as the locus flanked by SEQ ID NO:32 and SEQ ID NO:33), the gpdIIA locus (defined as the locus flanked by SEQ ID NO:34 and SEQ ID NO:35); and the gpdIIB locus (defined as the locus flanked by SEQ ID NO:38 and SEQ ID NO:39). Other suitable integration loci may be determined one of skill in the art. Furthermore, one of skill in the art would recognize how to use sequences to design primers to verify correct gene integration at the chosen locus.

[0045] The recombinant cell may have one or more copies of a given exogenous nucleic acid sequence integrated in a host chromosome(s) and replicated together with the chromosome(s) into which it has been integrated. For example, the yeast cell may be transformed with nucleic acid construct including a polynucleotide sequence encoding for a polypeptide described herein and the polynucleotide sequence encoding for the polypeptide may be integrated in one or more copies in a host chromosome(s). The recombinant cell may include multiple copies (two or more) of a given polynucleotide sequence encoding a polypeptide described herein. The recombinant cell may have one, two, three, four, five, six, seven, eight, nine, ten, or more copies of a polynucleotide sequence encoding a polypeptide described herein integrated into the genome. The multiple copies of said polynucleotide sequence may all be incorporated at a single locus or may be incorporated at multiple loci.

[0046] The recombinant cells described herein are capable of producing arabitol and include an exogenous polynucleotide sequence encoding an arabitol 2-dehydrogenase (ARD2DH) enzyme. The exogenous polynucleotide sequence may be an exogenous ARD2DH gene.

[0047] An "arabitol 2-dehydrogenase gene" and an "ARD2DH gene" are used interchangeably herein and refer to any gene or polynucleotide that encodes a polypeptide with arabitol 2-dehydrogenase activity. As used herein "arabitol 2-dehydrogenase activity" refers to the ability to catalyze the conversation of D-ribulose and NADH or NADPH to D-arabitol and NAD^+ or $NADP^+$. Enzymes with arabitol 2-dehydrogenase may be characterized under Enzyme Classification 1.1.1.250. The ARD2DH gene may be derived from any suitable source. For example, the ARD2DH gene may be derived from *Beauveria bassiana, Pichia stipitis, Candida albicans, Kwoniella heveanensis, Candida maltosa*. The ARD2DH gene may encode a polypeptide with at least 50%, at least 60%, at least 70%, at least 80%, at least 85%, at least 90%, at least 95%, at least 97%, or at least 99% sequence identity to the amino acid sequence of at least one of SEQ ID NOs:1, 2, 3, 9, or 11. The ARD2DH gene may encode a polypeptide with at least 50%, at least 60%, at least 70%, at least 80%, at least 85%, at least 90%, at least 95%, at least 97%, or at least 99% sequence identity to the amino acid sequence of at least one of SEQ ID NOs:2, 3, 9, or 11.

[0048] The recombinant cell may comprise an exogenous polynucleotide that is or may be derived from a *Beauveria bassiana* ARD2DH gene encoding the amino acid of SEQ ID NO:1. The exogenous polynucleotide may encode an amino acid sequence with at least 50%, at least 60%, at least 70%, at least 80%, at least 85%, at least 90%, at least 95%, at least 97%, or at least 99% sequence identity to the amino acid sequence of SEQ ID NO:1.

[0049] The recombinant cell may comprise an exogenous polynucleotide that is or may be derived from a *Pichia stipitis* ARD2DH gene encoding the amino acid of SEQ ID NO:2. The exogenous polynucleotide may encode an amino acid sequence with at least 50%, at least 60%, at least 70%, at least 80%, at least 85%, at least 90%, at least 95%, at least 97%, or at least 99% sequence identity to the amino acid sequence of SEQ ID NO:2.

[0050] The recombinant cell may comprise an exogenous polynucleotide that is or may be derived from a *Candida albicans* ARD2DH gene encoding the amino acid of SEQ ID NO:3. The exogenous polynucleotide may encode an amino acid sequence with at least 50%, at least 60%, at least 70%, at least 80%, at least 85%, at least 90%, at least 95%, at least 97%, or at least 99% sequence identity to the amino acid sequence of SEQ ID NO:3.

[0051] The recombinant cell may comprise an exogenous polynucleotide that is or may be derived from a *Kwoniella heveanensis* ARD2DH gene encoding the amino acid of SEQ ID NO:9. The exogenous polynucleotide may encode an amino acid sequence with at least 50%, at least 60%, at least 70%, at least 80%, at least 85%, at least 90%, at least 95%, at least 97%, or at least 99% sequence identity to the amino acid sequence of SEQ ID NO:9.

[0052] The recombinant cell may comprise an exogenous polynucleotide that is or may be derived from a *Candida maltosa* ARD2DH gene encoding the amino acid of SEQ ID NO:11. The exogenous polynucleotide may encode an amino acid sequence with at least 50%, at least 60%, at least 70%, at least 80%, at least 85%, at least 90%, at least 95%, at least 97%, or at least 99% sequence identity to the amino acid sequence of SEQ ID NO:11.

[0053] The exogenous polynucleotides in the recombinant cells described herein may be under the control of a promoter. For example, the exogenous nucleic acid may be operably linked to a heterologous or artificial promoter. Suitable promoters are known and described in the art. Promoters may include, but are not limited to, pyruvate decarboxylase promoter (PDC), translation elongation factor 2 promoter (TEF2), SED1, alcohol dehydrogenase 1A promoter (ADH1), hexokinase 2 promoter (HXK2), FLO5 promoter, pyruvate kinase 1 promoter (PYK1p; SEQ ID NO:49), 6-phosphogluconate dehydrogenase promoter (6PGDp; SEQ ID NO:40), glyceraldehyde-3-phosphate dehydrogenase promoter (TDH3p; SEQ ID NO:42), translational elongation factor 1 promoter (TEFp; SEQ ID NO:43), modified TEFp (SEQ ID NO:41), phosphoglucomutase 1 promoter (PGM1p; SEQ ID NO:44), 3-phosphoglycerate kinase promoter (PGK1p; SEQ ID NO:45), enolase promoter (ENO1p; SEQ ID NO:46), asparagine synthetase promoter (ASNSp; SEQ ID NO:47), 50S ribosomal protein L1 promoter (RPLAp; SEQ ID NO:48), and RPL16B (SEQ ID NO:50).

[0054] The exogenous nucleic acids in the recombinant cells described herein may be under the control of a terminator. For example, the exogenous nucleic acid may be operably linked to a heterologous or artificial terminator. Suitable terminators are known and described in the art. Terminators may include, but are not limited to, GAL10 terminator, PDC terminator, transaldolase terminator (TAL) 6PGD terminator (6PGDt; SEQ ID NO:51); ASNS terminator (ASNSt; SEQ ID NO:52); ENO1 terminator (ENO1t; SEQ ID NO:53); hexokinase 1 terminator (HXK1t; SEQ ID NO:54); PGK1 terminator (PGK1t; SEQ ID NO:55); PGM1 terminator (PGM1t; SEQ ID NO:56); PYK1 terminator (PYK1t; SEQ ID NO:57); RPLA terminator (RPLAt: SEQ ID NO:58); transaldolase 1 terminator (TAL1t; SEQ ID NO:59); TDH3 terminator (TDH3t; SEQ ID NO:60); translation elongation factor 2 terminator (TEF2t; SEQ ID NO:61); and triosephosphate isomerase 1 terminator (TPI1t; SEQ ID NO:62).

[0055] A promoter or terminator is "operably linked" to a given polynucleotide (e.g., a gene) if its position in the genome or expression cassette relative to said polynucleotide is such that the promoter or terminator, as the case may be, performs its transcriptional control function.

[0056] The polypeptides described herein may be provided as part of a construct. As used herein, the term "construct" refers to recombinant polynucleotides including, without limitation, DNA and RNA, which may be single-stranded or double-stranded and may represent the sense or the antisense strand. Recombinant polynucleotides are polynucleotides formed by laboratory methods that include polynucleotide sequences derived from at least two different natural sources or they may be synthetic. Constructs thus may include new modifications to endogenous genes introduced by, for example, genome editing technologies. Constructs may also include recombinant polynucleotides created using, for example, recombinant DNA methodologies. The construct may be a vector including a promoter operably linked to the polynucleotide encoding a polypeptide as described herein. As used herein, the term "vector" refers to a polynucleotide capable of transporting another polynucleotide to which it has been linked. The vector may be a plasmid, which refers to a circular double-stranded DNA loop into which additional DNA segments may be integrated.

[0057] The disclosure also provides fermentation methods for the production of arabitol using the recombinant cells described herein. The fermentation methods include the step of fermenting a substrate using the genetically engineered yeasts described herein to product arabitol. The fermentation method can include additional steps, as would be understood by a person skilled in the art. Non-limiting examples of additional process steps include maintaining the temperature of the fermentation broth within a predetermined range, adjusting the pH during fermentation, and isolating the arabitol from the fermentation broth. The fermentation process may be a fully aerobic process.

[0058] The fermentation method can be run using a suitable fermentation substrate. The substrate of the fermentation method can include glucose, sucrose, galactose, mannose, molasses, xylose, fructose, hydrolysates of starch, lignocellulosic hydrolysates, or a combination thereof. One skilled in the art will recognize what fermentation substrate is suitable for a given fermentation organism and system.

[0059] The fermentation process can be run under various conditions. The fermentation temperature, i.e., the temperature of the fermentation broth during processing, may be ambient temperature. Alternatively, or additionally, the fermentation temperature may be maintained within a predetermined range. For example, the fermentation temperature can be maintained in the range of 25° C. to 45° C., 30° C. to 40° C., or 32° C. to 37° C., preferably about 35° C. However, a skilled artisan will recognize that the fermentation temperature is not limited to any specific range or temperature recited herein and may be modified as appropriate.

[0060] The fermentation process can be run within certain oxygen uptake rate (OUR) ranges. The volumetric OUR of the fermentation process can be in the range of 0.5 to 40, 1 to 35, 2 to 30, 3 to 25, 4 to 20, or 5 to 15 mmol $O_2/(L \cdot h)$. In some embodiments, the specific OUR can be in the range of 0.05 to 10, 0.1 to 8, 0.15 to 5, 0.2 to 1, or 0.3 to 0.75 mmol O_2/(g cell dry weight·h). However, the volumetric or specific OURs of the fermentation process are not limited to any specific rates or ranges recited herein.

[0061] The fermentation process can be run at various cell concentrations. In some embodiments, the cell dry weight at the end of fermentation can be 5 to 40, 8 to 30, or 10 to 20 g cell dry weight/L. Further, the pitch density or pitching rate of the fermentation process can vary. In some embodiments, the pitch density can be 0.05 to 11, 0.1 to 10, or 0.25 to 8 g cell dry weight/L.

[0062] The initial dextrose concentration of the fermentation may be at least 100, 200, 250, 300, 350, or at least 400 g/L dextrose. The initial dextrose concentration may be between 100 to 400, 150 to 350, or 250 to 325 g/L.

[0063] The fermentation process can be associated with various characteristics, such as, but not limited to, fermentation production rate, pathway fermentation yield, final titer, and peak fermentation rate. These characteristics can be affected by the selection of the yeast and/or genetic modification of the yeast used in the fermentation process. These characteristics can be affected by adjusting the fermentation process conditions. These characteristics can be adjusted via a combination of yeast selection or modification and the selection of fermentation process conditions.

[0064] The final arabitol titer of the process may be at least 0.2, 0.5, 0.75, 1.0, 1.5, or 2.0 g/L.

[0065] The fermentation process can be run as a dextrose-fed batch. Further, the fermentation process can be a batch process, continuous process, or semi-continuous process, as would be understood by a person skilled in the art.

EXAMPLES

[0066] The invention is further described in detail by reference to the following experimental examples. These examples are provided for purposes of illustration only and are not intended to be limiting unless otherwise specified. Thus, the invention should in no way be construed as being limited to the following examples, but rather should be construed to encompass any and all variations which become evident as a result of the teaching provided herein.

Example 1—D-Arabitol 2-Dehydrogenase Diversity

[0067] 509 candidate D-arabitol 2-dehydrogenase (ARD2DH) enzyme sequences were selected from the Uniprot database and analyzed. FIG. **2** demonstrates the sequences diversity for this set of sequences. This set is diverse, with 35% of the sequences having no homolog with more than 75% sequence identity. Eleven of these sequences were chosen for further characterization in vivo.

Example 2—Genetically Modified *Moniliella pollinis* Strains

[0068] Strain 1-1 is the *Moniliella pollinis* host strain "*Moniliella tomentosa* var *pollinis* TCV364" described in U.S. Pat. No. 6,440,712, which is incorporated herein by reference in its entirety, and deposited under the Budapest Treaty at BCCM/MUCL (Belgian Coordinated Collections of Micro-organisms/Mycothéque de l'Université Catholique de Louvain by *Eridania* Béghin Say, Vilvoorde R&D Centre, Havenstraat 84, B-1800 Vilvoorde) on Mar. 28, 1997 under number MUCL40385. Table 2 below lists various *Moniliella pollinis* strains, including information on the parent strain, the sequence with which the parent strain was transformed, and characterizations of the expression cassette (s) contained on the transformed sequence. Each "ARD2DH Homolog Expression Cassette" contained, in order, a 3' Zeocin resistance gene bipartite fragment (SEQ ID NO:65), the TEF2 terminator (SEQ ID NO:61), an Mp6PGD promoter (SEQ ID NO:40), a gene encoding the indicated ARD2DH homolog (one of SEQ ID NOs:1-11), an Mp6PGD terminator (SEQ ID NO:51), and a 3' ER3 flanking sequence (SEQ ID NO:25). Each "Selectable Marker Cassette" contained, in order, a 5' ER3 flanking sequence (SEQ ID NO:24), and a 5' Zeocin resistance gene bipartite fragment (SEQ ID NO:66).

[0069] Upon bipartite transformation with both the ARD2DH Homolog Expression Cassette and the Selectable Marker Cassette, the two cassettes recombine for integration of both the nucleotide sequence encoding the ARD2DH homolog and the Zeocin selectable marker at the ER3 locus.

[0070] The indicated *Moniliella pollinis* parent strain was transformed with the indicated sequence(s) by first protoplasting the parent strain by adding an enzyme mixture containing 0.6M $MgSO_4$, 7.5 g/L driselase, and 12.5 g/L *Trichoderma harzianum* lysing enzyme to a mycelial pellet of the parent strain. Protoplasts were then pelleted, washed with 0.6M $MgSO_4$, and resuspended in STC medium (0.6M sucrose, 50 mM CaCl2), 10 mM Tris-HCl, pH 7.5). Bipartite transformations were prepared by adding 100 μg single stranded salmon sperm DNA and 1.5 to 5 μg each of the 5' and 3' DNA transformation fragments (3-10 μg total; see Table 2 for list of fragments) to approximately 200 μL protoplast mixture (10^8 cells/mL). 1 mL 50% PEG in STC medium was then added to the salmon sperm DNA, transformation DNA, and protoplast mixture and the resulting combination was incubated for 15 minutes at room temperature. Following incubation, recovery broth (0.4M sucrose, 1 g/L yeast extract, 1 g/L malt extract, 10 g/L glucose, pH 4.5) was added to the mixture and incubated at 27° C., 100 rpm, for 16 to 24 hours. Following the incubation, protoplasts were pelleted by centrifugation and resuspended in 1 mL PBS.

[0071] The resuspended protoplasts were plated on PDA+100 mg/L zeocin selection plates and incubated at 30-35° C. for at least 2-4 days until transformants grow. Resulting transformants were evaluated by colony PCR for integration of the indicated sequence. A PCR verified isolate was then designated as the indicated strain number. In some instances, more than one PCR verified isolate, e.g., "sister" isolates, are indicated by letters following the strain number. For example, strain 1-2 has 4 sister isolates, strains 1-2a, 1-2b, 1-2c, and 1-2d.

[0072] For example, Strain 1-1 was transformed with SEQ ID NO:12 and SEQ ID NO:13. SEQ ID NO: 12 contains i) 5' flanking DNA for targeted chromosomal integration into the ER3 locus (SEQ ID NO:24), and ii) a 5' portion of the Zeocin selectable marker (SEQ ID NO:66). SEQ ID NO:13 contains i) a 3' portion of the Zeocin selectable marker (SEQ ID NO:65), ii) the TEF2 terminator (SEQ ID NO:61), iii) an Mp6PGD promoter (SEQ ID NO:40), iv) a gene encoding the *Beauveria bassiana* ARD2DH homolog of SEQ ID NO:1, v) an Mp6PGD terminator (SEQ ID NO:51), and vi) a 3' ER3 flanking sequence (SEQ ID NO:25). Transformants were selected on PDA+100 mg/L zeocin selection plates and incubated at 30-35° C. for at least 2 days until transformants grow. Resulting transformants were streaked for single colony isolation on PDA+zeocin plates and single colonies were selected. Selected colonies were evaluated by colony PCR for presence of the indicated sequence. PCR verified isolates were designated strains 1-2a, 1-2b, 1-2c, and 1-2d.

TABLE 2

		ARD2DH Homolog Expression Cassette			Selectable Marker Cassette	
Strain	Parent Strain	Transformation SEQ ID NO:	Polypeptide Source Organism	Encoded Polypeptide SEQ ID NO:	Selection Marker	Transformation SEQ ID NO:
1-2a-d	1-1	13	*Beauveria bassiana*	1	Zeocin	12
1-3a-f	1-1	14	*Pichia stipitis*	2	Zeocin	12
1-4a-f	1-1	15	*Candida albicans*	3	Zeocin	12
1-5a-g	1-1	16	*Verticillium dahliae*	4	Zeocin	12
1-6a-f	1-1	17	*Fusarium proliferatum*	5	Zeocin	12
1-7a-e	1-1	18	*Aspergillus awamori*	6	Zeocin	12
1-8a-g	1-1	19	*Colletotrichum shisoi*	7	Zeocin	12
1-9a-c	1-1	20	*Ophiostoma piceae*	8	Zeocin	12
1-10a-h	1-1	21	*Kwoniella heveanensis*	9	Zeocin	12
1-11a-f	1-1	22	*Pichia kudriavzevii*	10	Zeocin	12
1-12a-g	1-1	23	*Candida maltosa*	11	Zeocin	12

Example 3—Shake Flask Fermentation Assay

[0073] Strains 1-1, 1-2a-d, 1-3a-f, 1-4a-f, 1-5a-g, 1-6a-f, and 1-7a-e (outlined in Table 2 above), were run in shake flasks to assess xylitol, erythritol, ribitol, glycerol, arabitol, and ethanol production and glucose consumption. As indicated in the tables below, some strains were run in duplicate.

[0074] Strains were streaked out for biomass growth on YPD plates (bacteriological peptone 20 g/L, yeast extract 10 g/L, glucose 20 g/L, and agar 15 g/L) and incubated at 30° C. for 48-72 hours. Cells from the incubated YPD plates were scraped into 40 mL rich medium (170 g/L glucose, 10 g/L yeast extract) in a 250 mL baffled flask. Cells were incubated at 30° C. and 250 rpm until the optical density (OD600) reached 15-20 to form the seed culture. Optical density is measured at a wavelength of 600 nm with a 1 cm path length cuvette using a model Genesys20 spectrophotometer (Thermo Scientific). The seed culture reached an OD600 between 15-20 in about 32-50 hours.

[0075] A 250 ml non-baffled flask containing production medium (Table 3) and antifoam CF-32 was inoculated with the seed culture to form the production culture with a starting OD600 of about 0.4 (approximately 0.8 mL of the seed culture). The production culture was incubated at 35° C. and 250 rpm. Samples were taken from the production culture after 72 and 96 hours of incubation. Samples were analyzed for glucose, ribitol, xylitol, erythritol, glycerol, arabitol, and ethanol by high performance liquid chromatography with refractive index detector. Fermentation results are reported in Tables C and D and FIGS. **3** and **4**. The expression of either the *Beauveria bassiana* ARD2DH homolog (SEQ ID NO:1), the *Pichia stipitis* ARD2DH homolog (SEQ ID NO:2), or the *Candida albicans* ARD2DH homolog (SEQ ID NO:3) in *Moniliella pollinis* resulted in production of arabitol at levels above that produced in the parent strain 1-1.

TABLE 3

Production Medium	
Component	Concentration (units)
Glucose	300 (g/L)
KH2PO4	1.27 (g/L)
(NH4)2HPO4	0.13 (g/L)
(NH4)2SO4	1.80 (g/L)
Urea	2.85 (g/L)
Citric Acid	200.0 (mg/L)
MgSO4—7H2O	690.2 (mg/L)
FeSO4—7H2O	32.35 (mg/L)
ZnSO4—7H2O	4.40 (mg/L)
MnSO4—H2O	0.53 (mg/L)
CuSO4—5H2O	0.45 (mg/L)
CaCl2—2H2O	91.60 (mg/L)
Na2SO4	154.50 (mg/L)
Thiamine-HCl	10 (mg/L)
Choline-Cl	100 (mg/L)
Antifoam CF-32	1.00 (g/L)

TABLE 4

Shake Flask Results from HPLC analysis							
	Time	Fermentation Broth Analyte (g/L)					
Strain	point	Glucose	Ribitol	Xylitol	Erythritol	Glycerol	Ethanol
1-2a	72 h	138.974	0.952	n.a.	42.458	36.162	12.128
1-2a	72 h	154.818	0.877	n.a.	39.143	35.189	12.830
1-2b	72 h	147.230	0.858	n.a.	40.215	31.736	10.654
1-2c	72 h	171.870	1.066	n.a.	40.236	21.374	9.887
1-2d	72 h	123.766	0.840	n.a.	55.664	33.503	16.918
1-3a	72 h	53.969	1.232	n.a.	64.291	74.409	10.532
1-3b	72 h	197.030	0.702	n.a.	28.731	27.809	6.502
1-3b	72 h	187.903	0.560	n.a.	31.747	31.654	4.260
1-3c	72 h	130.937	0.997	n.a.	50.702	37.837	16.380

TABLE 4-continued

Shake Flask Results from HPLC analysis							
	Time	Fermentation Broth Analyte (g/L)					
Strain	point	Glucose	Ribitol	Xylitol	Erythritol	Glycerol	Ethanol
1-3d	72 h	142.550	0.999	n.a.	42.669	30.431	13.946
1-3d	72 h	96.848	1.325	n.a.	64.914	39.370	14.251
1-3e	72 h	73.658	1.242	n.a.	66.444	47.364	17.622
1-3f	72 h	121.154	1.021	n.a.	48.641	31.506	16.467
1-4a	72 h	151.311	0.651	n.a.	39.050	37.208	11.610
1-4a	72 h	156.634	0.473	n.a.	37.185	34.785	13.796
1-4b	72 h	85.931	0.814	n.a.	59.658	56.796	12.817
1-4b	72 h	97.442	0.715	n.a.	56.453	44.355	17.953
1-4c	72 h	94.773	0.941	n.a.	61.601	46.229	13.104
1-4d	72 h	111.870	0.832	n.a.	55.710	37.555	17.192
1-4d	72 h	139.961	0.941	n.a.	49.182	36.785	14.520
1-4e	72 h	172.284	0.516	n.a.	32.121	28.667	10.100
1-4f	72 h	74.533	1.349	n.a.	59.577	62.783	13.522
1-5a	72 h	202.443	0.596	0.056	23.629	14.945	14.594
1-5a	72 h	187.870	0.618	0.051	26.423	15.886	16.484
1-5b	72 h	112.036	1.256	0.049	57.326	36.291	13.694
1-5c	72 h	87.601	1.267	0.051	66.859	44.340	13.306
1-5d	72 h	144.311	1.036	0.024	45.567	30.897	16.153
1-5e	72 h	169.817	0.623	0.032	32.422	31.399	13.399
1-5f	72 h	111.758	1.115	0.037	53.894	35.848	14.645
1-5g	72 h	62.216	2.007	0.078	64.188	68.604	11.441
1-6a	72 h	95.951	0.841	0.045	60.461	35.540	18.394
1-6b	72 h	161.823	0.867	0.053	46.684	19.524	13.461
1-6c	72 h	333.753	0.036	n.a.	1.621	0.988	2.392
1-6d	72 h	60.287	1.324	n.a.	72.479	76.702	6.809
1-6d	72 h	72.865	1.176	n.a.	67.559	58.915	12.283
1-6e	72 h	165.974	0.863	0.062	40.876	17.582	17.366
1-6f	72 h	146.021	10.838	0.046	43.472	30.099	13.857
1-7a	72 h	175.392	0.689	0.024	34.772	23.845	12.074
1-7b	72 h	142.213	0.739	0.041	42.201	23.216	20.619
1-7d	72 h	126.273	1.435	0.038	50.768	36.879	15.641
1-7e	72 h	41.556	2.501	0.065	79.599	70.089	11.542
1-7e	72 h	32.295	2.501	0.057	76.289	74.286	10.538
1-7f	72 h	213.204	0.704	n.a.	27.133	22.063	11.927
1-1	72 h	79.170	1.231	0.040	65.958	47.596	12.070
1-1	72 h	97.698	0.991	n.a.	59.115	35.557	18.380

TABLE 5

Shake Flask Results from HPIC Analysis									
	Time	Analyte (g/L)				Time	Analyte (g/L)		
Strain	Point	Xylitol	Ribitol	Arabitol	Strain	Point	Xylitol	Ribitol	Arabitol
1-2a	72 h	n.a.	1.115	0.052	1-5e	72 h	n.a.	0.857	0.039
1-2a	72 h	n.a.	2.100	0.074	1-5f	72 h	n.a.	1.370	0.052
1-2b	72 h	n.a.	1.135	0.048	1-5g	72 h	n.a.	1.934	0.069
1-2c	72 h	n.a.	1.324	0.042	1-6a	72 h	n.a.	0.966	0.055
1-2d	72 h	n.a.	0.988	0.055	1-6b	72 h	n.a.	0.905	0.046
1-3a	72 h	n.a.	1.424	0.759	1-6c	72 h	n.a.	0.043	n.a.
1-3b	72 h	n.a.	0.724	0.209	1-6d	72 h	n.a.	1.574	0.067
1-3b	72 h	n.a.	0.694	0.201	1-6d	72 h	n.a.	1.288	0.056
1-3c	72 h	n.a.	1.205	0.611	1-6e	72 h	n.a.	1.183	0.051
1-3d	72 h	n.a.	1.293	0.465	1-6f	72 h	n.a.	1.009	0.047
1-3d	72 h	n.a.	1.589	0.659	1-7a	72 h	n.a.	0.854	0.049
1-3e	72 h	n.a.	1.639	0.708	1-7b	72 h	n.a.	0.992	0.052
1-3f	72 h	n.a.	1.189	0.596	1-7d	72 h	n.a.	1.629	0.051
1-4a	72 h	n.a.	0.750	0.211	1-7e	72 h	n.a.	2.591	0.072
1-4a	72 h	n.a.	0.750	0.212	1-7e	72 h	n.a.	2.587	0.071
1-4b	72 h	n.a.	0.993	0.273	1-7f	72 h	n.a.	0.706	0.049
1-4b	72 h	n.a.	0.898	0.264	1-1	72 h	n.a.	1.292	0.058
1-4c	72 h	n.a.	1.034	0.436	1-1	72 h	n.a.	1.097	0.053
1-4d	72 h	n.a.	0.930	0.287	1-2a	96 h	n.a.	1.941	0.137
1-4d	72 h	n.a.	0.997	0.297	1-2a	96 h	n.a.	1.941	0.312
1-4e	72 h	n.a.	0.832	0.336	1-2b	96 h	n.a.	2.288	0.055
1-4f	72 h	n.a.	1.494	0.575	1-2c	96 h	n.a.	2.670	0.047
1-5a	72 h	n.a.	1.227	0.054	1-2d	96 h	n.a.	1.476	0.156

TABLE 5-continued

Shake Flask Results from HPIC Analysis									
	Time	Analyte (g/L)				Time	Analyte (g/L)		
Strain	Point	Xylitol	Ribitol	Arabitol	Strain	Point	Xylitol	Ribitol	Arabitol
1-5a	72 h	n.a.	1.293	0.052	1-3a	96 h	n.a.	1.529	0.822
1-5b	72 h	n.a.	1.317	0.053	1-3b	96 h	n.a.	1.739	0.509
1-5c	72 h	n.a.	1.321	0.057	1-3b	96 h	n.a.	1.617	0.397
1-5d	72 h	n.a.	1.243	0.054	1-3c	96 h	n.a.	1.893	1.124
1-3d	96 h	n.a.	2.449	0.938	1-5e	96 h	n.a.	1.558	0.011
1-3d	96 h	n.a.	2.088	0.956	1-5f	96 h	n.a.	2.227	0.032
1-3e	96 h	n.a.	1.705	0.803	1-5g	96 h	n.a.	2.126	0.062
1-3f	96 h	n.a.	1.784	1.084	1-6a	96 h	n.a.	1.185	0.065
1-4a	96 h	n.a.	1.291	0.397	1-6b	96 h	n.a.	1.828	0.068
1-4a	96 h	n.a.	1.341	0.395	1-6c	96 h	n.a.	0.022	n.a.
1-4b	96 h	n.a.	1.170	0.335	1-6d	96 h	n.a.	1.641	0.054
1-4b	96 h	n.a.	1.039	0.314	1-6d	96 h	n.a.	1.535	0.050
1-4c	96 h	n.a.	1.150	0.500	1-6e	96 h	n.a.	2.283	0.068
1-4d	96 h	n.a.	1.420	0.446	1-6f	96 h	n.a.	1.795	0.044
1-4d	96 h	n.a.	1.775	0.525	1-7a	96 h	n.a.	1.839	0.043
1-4e	96 h	n.a.	1.618	0.619	1-7b	96 h	n.a.	1.878	0.071
1-4f	96 h	n.a.	1.691	0.709	1-7d	96 h	n.a.	2.289	0.040
1-5a	96 h	n.a.	1.730	0.013	1-7e	96 h	n.a.	2.714	0.057
1-5a	96 h	n.a.	1.636	0.015	1-7e	96 h	n.a.	2.709	0.056
1-5b	96 h	n.a.	1.853	0.046	1-7f	96 h	n.a.	1.468	0.059
1-5c	96 h	n.a.	1.492	0.044	1-1	96 h	n.a.	1.551	0.048
1-5d	96 h	n.a.	2.729	0.064	1-1	96 h	n.a.	1.430	0.045

Example 4—Shake Flask Fermentation Assay

[0076] Strains 1-1, (outlined in Table 2 above), were run in shake flasks to assess xylitol, erythritol, ribitol, glycerol, arabitol, and ethanol production and glucose consumption. As indicated in the tables below, some strains were run in duplicate.

[0077] Strains were streaked out for biomass growth on YPD plates (bacteriological peptone 20 g/L, yeast extract 10 g/L, glucose 20 g/L, and agar 15 g/L) and incubated at 30° C. for 48-72 hours. Cells from the incubated YPD plates were scraped into 40 mL rich medium (170 g/L glucose, 10 g/L yeast extract) in a 250 mL baffled flask. Cells were incubated at 30° C. and 250 rpm until the optical density (OD600) reached 15-20 to form the seed culture. Optical density is measured at a wavelength of 600 nm with a 1 cm path length cuvette using a model Genesys20 spectrophotometer (Thermo Scientific). The seed culture reached an OD600 between 15-20 in about 32-50 hours.

[0078] A 250 ml non-baffled flask containing production medium (Table 3) and antifoam CF-32 was inoculated with the seed culture to form the production culture with a starting OD600 of about 0.4 (approximately 0.8 mL of the seed culture). The production culture was incubated at 35° C. and 250 rpm. Samples were taken from the production culture after 72 and 96 hours of incubation. Samples were analyzed for glucose, ribitol, xylitol, erythritol, glycerol, arabitol, and ethanol by high performance liquid chromatography with refractive index detector. Fermentation results are reported in Tables 6 and 7 and FIGS. **5** and **6**. The expression of either the *Kwoniella heveanensis* ARD2DH homolog (SEQ ID NO:9) or the *Candida maltosa* ARD2DH homolog (SEQ ID NO: 11) in *Moniliella pollinis* resulted in production of arabitol at levels above that produced in the parent strain 1-1.

TABLE 6

Shake Flask Results from HPLC Analysis							
	Time	Fermentation Broth Analyte (g/L)					
Strain	Point	Glucose	Ribitol	Xylitol	Erythritol	Glycerol	Ethanol
1-8a	72 h	258.093	0.345	n.a.	14.677	15.765	3.155
1-8b	72 h	158.312	0.622	n.a.	35.310	34.187	13.640
1-8c	72 h	122.825	0.485	n.a.	45.629	39.945	18.591
1-8c	72 h	143.231	0.451	n.a.	40.329	37.153	17.584
1-8d	72 h	158.401	0.905	n.a.	48.649	44.640	6.130
1-8e	72 h	85.216	0.549	0.071	38.519	71.841	11.367
1-8f	72 h	129.561	0.388	0.091	30.789	40.494	19.172
1-8g	72 h	142.656	0.936	n.a.	42.320	37.962	15.272
1-9a	72 h	174.899	0.615	n.a.	33.936	24.944	17.186
1-9a	72 h	93.589	1.129	n.a.	66.802	68.045	8.835
1-9b	72 h	183.544	0.612	n.a.	34.047	13.051	16.074
1-9b	72 h	185.236	0.684	n.a.	38.433	14.545	17.091
1-9c	72 h	139.892	1.088	n.a.	39.281	41.351	14.757
1-9c	72 h	94.137	1.803	n.a.	55.768	64.523	14.119
1-10a	72 h	124.205	0.469	n.a.	50.087	48.748	17.194

TABLE 6-continued

Shake Flask Results from HPLC Analysis

Strain	Time Point	Fermentation Broth Analyte (g/L) Glucose	Ribitol	Xylitol	Erythritol	Glycerol	Ethanol
1-10b	72 h	83.845	0.950	n.a.	76.573	48.320	15.627
1-10b	72 h	74.566	0.908	n.a.	74.046	51.312	15.903
1-10c	72 h	111.100	0.939	n.a.	57.084	35.657	18.351
1-10d	72 h	110.100	0.826	n.a.	54.148	51.672	18.477
1-10e	72 h	112.975	1.356	n.a.	65.674	17.845	28.830
1-10f	72 h	81.434	0.838	n.a.	65.170	57.235	11.768
1-10g	72 h	124.155	1.035	n.a.	54.286	17.195	26.670
1-10h	72 h	106.176	0.914	n.a.	56.241	47.372	11.749
1-11a	72 h	135.914	1.158	n.a.	54.830	39.250	13.827
1-11b	72 h	83.434	0.507	0.009	53.177	55.997	15.170
1-11c	72 h	174.592	0.487	n.a.	31.719	17.421	19.345
1-11d	72 h	109.258	0.918	n.a.	64.784	41.905	14.817
1-11e	72 h	60.799	1.058	n.a.	70.074	66.643	11.627
1-11f	72 h	240.070	0.417	n.a.	24.313	9.815	12.863
1-12a	72 h	140.680	0.585	n.a.	47.631	25.265	14.771
1-12b	72 h	142.136	0.638	n.a.	46.313	31.604	13.266
1-12c	72 h	140.644	0.697	n.a.	55.415	33.160	18.373
1-12d	72 h	163.982	0.764	0.149	33.370	40.143	7.184
1-12e	72 h	133.482	0.575	n.a.	56.197	26.830	19.126
1-12e	72 h	116.895	0.690	n.a.	59.420	27.340	15.744
1-12f	72 h	152.551	0.580	n.a.	43.082	23.114	18.110
1-12g	72 h	76.438	0.901	n.a.	76.386	42.281	16.117
1-1	72 h	71.718	1.154	n.a.	78.372	43.459	14.480
1-1	72 h	122.662	0.999	n.a.	59.657	34.594	15.310
1-8a	96 h	125.905	1.176	n.a.	52.441	33.086	8.894
1-8b	96 h	89.242	1.067	0.017	52.557	33.181	9.580
1-8c	96 h	80.668	0.817	0.012	72.887	41.899	14.091
1-8c	96 h	83.440	0.881	0.009	75.516	38.788	14.792
1-8d	96 h	71.801	1.648	n.a.	91.007	64.680	2.349
1-8e	96 h	62.620	0.645	0.094	43.124	75.463	6.033
1-8f	96 h	95.637	0.611	0.156	44.765	43.390	13.175
1-8g	96 h	69.125	1.704	n.a.	79.129	41.034	11.105
1-9a	96 h	99.365	1.122	n.a.	59.673	29.177	17.728
1-9a	96 h	25.113	1.464	n.a.	97.002	83.835	3.219
1-9b	96 h	125.942	1.195	n.a.	62.046	14.589	17.479
1-9b	96 h	126.024	1.254	n.a.	67.902	15.353	18.603
1-9c	96 h	86.244	2.428	n.a.	71.410	58.726	13.919
1-9c	96 h	36.964	2.817	n.a.	77.083	69.199	9.929
1-10a	96 h	86.482	0.652	n.a.	70.213	48.729	13.014
1-10b	96 h	54.649	1.263	n.a.	90.452	48.661	10.267
1-10b	96 h	55.176	1.148	n.a.	90.379	56.771	12.657
1-10c	96 h	45.006	1.620	n.a.	95.711	33.979	9.173
1-10d	96 h	35.704	1.564	n.a.	94.157	55.305	15.210
1-10e	96 h	64.952	1.699	n.a.	82.159	21.388	28.231
1-10f	96 h	44.240	1.124	n.a.	89.910	65.906	9.127
1-10g	96 h	67.965	1.615	n.a.	85.709	17.156	31.461
1-10h	96 h	56.950	1.354	n.a.	90.701	56.190	7.182
1-11a	96 h	40.113	1.720	n.a.	87.543	49.303	11.521
1-11b	96 h	44.750	0.818	0.095	72.836	60.261	10.697
1-11c	96 h	111.065	0.832	n.a.	53.207	17.391	22.468
1-11d	96 h	76.768	1.291	0.112	87.903	40.498	12.141
1-11e	96 h	47.106	1.241	0.102	81.480	68.506	9.020
1-11f	96 h	160.022	1.167	n.a.	60.755	13.121	16.294
1-12a	96 h	85.980	1.169	n.a.	95.368	25.823	10.069
1-12b	96 h	73.770	1.223	n.a.	93.507	35.887	7.098
1-12c	96 h	65.225	1.155	n.a.	95.494	27.027	14.818
1-12d	96 h	134.939	1.119	0.213	45.002	56.199	5.436
1-12e	96 h	48.987	0.945	n.a.	98.830	20.934	18.023
1-12e	96 h	54.041	1.209	n.a.	107.092	25.727	12.807
1-12f	96 h	71.820	1.190	n.a.	96.085	21.648	20.556
1-12g	96 h	61.693	1.076	n.a.	87.127	41.031	11.967
1-1	96 h	64.454	1.436	0.091	91.280	45.617	12.050
1-1	96 h	57.705	1.521	n.a.	96.210	30.674	7.946

TABLE 7

Shake Flask Results from HPIC Analysis				
		Fermentation Broth Analyte (g/L)		
Strain	Time Point	Xylitol	Ribitol	Arabitol
1-8a	72 h	n.a.	0.359	0.074
1-8b	72 h	0.214	0.869	0.109
1-8c	72 h	0.272	0.832	0.102
1-8c	72 h	0.230	0.761	0.096
1-8d	72 h	n.a.	1.037	0.094
1-8e	72 h	0.327	0.874	0.109
1-8f	72 h	0.288	0.589	0.094
1-8g	72 h	n.a.	0.915	0.098
1-9a	72 h	0.169	0.723	0.085
1-9a	72 h	n.a.	1.112	0.120
1-9b	72 h	0.132	0.747	0.031
1-9b	72 h	0.135	0.769	0.031
1-9c	72 h	n.a.	1.367	0.091
1-9c	72 h	n.a.	1.959	0.108
1-10a	72 h	n.a.	0.587	0.286
1-10b	72 h	n.a.	1.078	0.803
1-10b	72 h	n.a.	1.046	0.833
1-10c	72 h	0.186	1.038	0.519
1-10d	72 h	n.a.	1.003	0.623
1-10e	72 h	n.a.	1.754	0.091
1-10f	72 h	n.a.	1.016	0.816
1-10g	72 h	0.181	1.108	0.137
1-10h	72 h	n.a.	0.955	0.319
1-11a	72 h	n.a.	1.090	0.105
1-11b	72 h	0.288	0.804	0.102
1-11c	72 h	0.160	0.567	0.066
1-11d	72 h	n.a.	0.955	0.137
1-11e	72 h	n.a.	1.029	0.110
1-11f	72 h	n.a.	0.391	0.057
1-12a	72 h	0.289	0.846	0.615
1-12b	72 h	0.252	0.762	0.446
1-12c	72 h	0.271	0.909	0.701
1-12d	72 h	0.273	1.110	0.426
1-12e	72 h	0.277	0.849	0.653
1-12e	72 h	0.317	1.001	0.724
1-12f	72 h	n.a.	0.722	0.517
1-12g	72 h	0.378	1.105	0.906
1-1	72 h	n.a.	1.216	0.126
1-1	72 h	n.a.	1.024	0.098
1-8a	96 h	0.212	1.317	0.113

TABLE 7-continued

Shake Flask Results from HPIC Analysis				
		Fermentation Broth Analyte (g/L)		
Strain	Time Point	Xylitol	Ribitol	Arabitol
1-8b	96 h	0.264	1.604	0.116
1-8c	96 h	0.319	1.408	0.092
1-8c	96 h	0.289	1.416	0.090
1-8d	96 h	n.a.	1.997	0.098
1-8e	96 h	0.295	1.122	0.081
1-8f	96 h	0.345	0.896	0.073
1-8g	96 h	0.268	1.894	0.108
1-9a	96 h	0.180	1.621	0.145
1-9a	96 h	n.a.	1.751	0.126
1-9b	96 h	0.114	1.642	n.a.
1-9b	96 h	0.125	1.596	n.a.
1-9c	96 h	0.296	3.095	0.060
1-9c	96 h	0.231	3.749	0.189
1-10a	96 h	0.277	0.942	0.573
1-10b	96 h	0.269	1.667	1.311
1-10b	96 h	0.270	1.535	1.198
1-10c	96 h	0.451	2.210	1.248
1-10d	96 h	0.347	2.260	1.228
1-10e	96 h	0.240	3.247	0.081
1-10f	96 h	0.331	1.541	1.171
1-10g	96 h	0.311	2.391	0.178
1-10h	96 h	n.a.	1.539	0.464
1-11a	96 h	n.a.	2.099	0.131
1-11b	96 h	0.388	1.206	0.082
1-11c	96 h	0.236	1.334	0.119
1-11d	96 h	n.a.	1.397	0.080
1-11e	96 h	n.a.	1.328	0.082
1-11f	96 h	0.225	1.259	0.029
1-12a	96 h	0.505	1.676	1.324
1-12b	96 h	0.516	1.670	1.209
1-12c	96 h	0.509	1.747	1.542
1-12d	96 h	0.289	1.759	0.688
1-12e	96 h	0.501	1.605	1.523
1-12e	96 h	0.528	1.798	1.535
1-12f	96 h	0.381	1.636	1.382
1-12g	96 h	0.526	1.491	1.383
1-1	96 h	0.257	1.509	0.096
1-1	96 h	n.a.	1.866	0.090

SEQUENCE LISTING

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Sequence total quantity: 73
SEQ ID NO: 1           moltype = AA  length = 361
FEATURE                Location/Qualifiers
source                 1..361
                       mol_type = protein
                       organism = Beauveria bassiana
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LPSDPVRRAG GQYVKPTLQT FSLDKKVGLV TGGARGLGLV IAQGMVYSGA DVALVDMNSK  120
LTQKLGGSFT SADIPTEEEA EKQTKNLVKA FMRENPNAAR VPKVTAHFSD VSDPASVDAC  180
IAEVIEQHGK IDNLVTSAGF TENYEAVNYP IDRLRKLWAV NIDGTYLFAT SVARHLMERK  240
APGSMVLIGS MSGSIVNVPQ PQTPYNVSKA GVRHLASSLA VEWADANIRV NCLSPGYMMT  300
ALTAKILDEN PELKKKWVSL IPMGKMGDPG DLMGPVIFLL SENSKYVNGI DLRVDGGYTL  360
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SEQ ID NO: 2           moltype = AA  length = 278
FEATURE                Location/Qualifiers
source                 1..278
                       mol_type = protein
                       organism = Pichia stipitis
SEQUENCE: 2
MDYSYANVVP NFRLDGRLAI ITGGSGGLAA VISRALLAQG ADVALIDMNL ERTKSAAKEV  60
LGWGEETLKG EHASAIGQVS AWSCNIGDAE AVDATFSSIN EHHGKIADLL INTAGYCENF  120
PAETYPATNA ESIMKVNGLG SFYVSQSFAR PLIQNNLRGS IILIGSMSGT IVNDPQPQCM  180
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YNMSKAGVIH LVRSLACEWA KYNIRVNTLS PGYILTPLTR NVISGHTEMK EAWESKIPMK  240
RMAEPKEFVG SILYLASETA SSYTTGHNLV VDGGYECW                          278

SEQ ID NO: 3            moltype = AA  length = 281
FEATURE                 Location/Qualifiers
source                  1..281
                        mol_type = protein
                        organism = Candida albicans
SEQUENCE: 3
MDSAYWSYDN IVPSFRLDGK LVILTGGSGG LAAVVSRALL AKGADVALVD MNLERTQQAA  60
RDVLQWGEEQ MKGKYESPIG QVSAWSCNIG DAEAVDLTFK AINEHHGKIS SVLVNTAGYA  120
ENFPAEEYPA KNAENLMKVN GLGSFYVSQA FARPLIQNNM TGSIILIGSM SGTIVNDPQP  180
QCMYNMSKAG VIHLARSLAC EWAKYNIRVN TLSPGYILTP LTRNVISGHT EMKTEWESKI  240
PMKRMAEPKE FVGSILYLAS ESASSYTTGH NLVVDGGYEC W                      281

SEQ ID NO: 4            moltype = AA  length = 361
FEATURE                 Location/Qualifiers
source                  1..361
                        mol_type = protein
                        organism = Verticillium dahliae
SEQUENCE: 4
MSSAKGPTSL SSDKILLLVA APPKQDWISR IESRYPGIEI RWVDCTSSGA LTTPDEIPAA  60
LWEDITIYSS WLTPRAELIP NVRFVQLSSA GADRWLDHAL YLNPDVVFCS ANGVHPPQIA  120
EWVIGTWLSH QHRFLHYHDS QREASWANRL AYPVEDSVGL RMGILGYGAI GRQCGRVAQA  180
LGMEVYAYTR SERADPASRK DDSYWVPGTG DPEGVLPARW FHGADVESIN AFLAQDLDLL  240
VVGLPLTKET DGILGAKQFD ILSKKKTFVS NIARGKLIDQ EALVAALEQG KIRGAALDVT  300
DPEPLPADHP LWKAPNVFLT PHVSGRSTAY WDRALAIFEQ NLERWSEGKP LINEINRTLH  360
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SEQ ID NO: 5            moltype = AA  length = 271
FEATURE                 Location/Qualifiers
source                  1..271
                        mol_type = protein
                        organism = Fusarium proliferatum
SEQUENCE: 5
MANPYPFQDK VIVVSGASRG TGLSLSRYLL VRGAKVSMAA TSEENLKKAV AGIEQDIPDA  60
KDRVIYFPTD VRNPDDVKAW IEGTVAKWGE LDGAANVAAK MNKCIHPIED LELGELKEVL  120
DVNVIGTFNS IKYEMKNMKP GGSIVNCGSQ QVKYASGNMG AYAASKNAIR GLSQSAAYEG  180
GAKYPKNPIR VNLLCPGCID TDMIRQPLHL PNGAGTWTMT EGDHLTSIIK RYSKPEEIAA  240
SIAFLLGDES RFMTKQEIYV DGGWMEANYV A                                 271

SEQ ID NO: 6            moltype = AA  length = 297
FEATURE                 Location/Qualifiers
source                  1..297
                        mol_type = protein
                        organism = Aspergillus awamori
SEQUENCE: 6
MATRSDCNVH GLEFREAKPS QGDPARPVAG LFRLDNRTII SISPGITGAT GFLGTTLAIA  60
ILESGGDVVC LDLASAPNSP DWNKVEAAAQ KHARQLSYYP LDVTDESGVE KTFTQFIPTL  120
RYPLKGIVAC AGLSRNGPAT EFPVSSFRRM LDINVTGTFL VAQAAAREML RTNTTGSMVF  180
VASMSGYVSN KGVDTAGYNA SKAAVHQLTR SLAAEWGSRV GMPLIRVNSL SPGYIRTAAT  240
AEALQKPGME SQWVGDNMLF RLSTADEFRA PVLFMLGDGS SFMTGADLRV DGGHCAW     297

SEQ ID NO: 7            moltype = AA  length = 324
FEATURE                 Location/Qualifiers
source                  1..324
                        mol_type = protein
                        organism = Colletotrichum shisoi
SEQUENCE: 7
MSRYLTRLRP LRLGPQLTLG KPFRLAVQPI TAPVAPIVGH QPKAFHASPK PLGPHITSPT  60
LTRVLPEFNL LGKVIVVSGG ARGLGLTQAE ALLEAGAIVH AIDRLPKPDP DFEKVALRAN  120
TELATSLTYH QVDVRDVPAL NEIVEGIANR HGRLDGLIAA AGIQQETPAL EYKAEDVDKM  180
MGVNVTGAFM TAQAVARQMV RLNTSGSIVL IASMSGTVAN RGLICPAYNA SKAAVLQLAR  240
NLAAEWGVHG IRVNTISPGY IVTAMVEALF ETYPERKVEW PKHNMLGRLS QPNEYRGAAV  300
FLLSDASSFM TGSDLRIDGG HCAW                                         324

SEQ ID NO: 8            moltype = AA  length = 373
FEATURE                 Location/Qualifiers
source                  1..373
                        mol_type = protein
                        organism = Ophiostoma piceae
SEQUENCE: 8
MSFSVARSVM SSARVASKAA PFARSALAKP AAFRAPVQSR AISVTSRLSA KNEVIKETEV  60
PVSIYSPDGK STSANPDQFN IPVKKGSAAA EASAEDLANA DVIPLDPKVY KSLPPTLQKM  120
SVMGKVIVVT GGARGLGNHM AKACAEAGAK ALVIFDANQE LGDEAAEELH RKSGIDVVFF  180
KVDVRDGNAI NAAVQQVVDI YGAPDVLVNS AGIADSNIKA ETYDPAMFRR LIDINLTGSF  240
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LMAQSVGRAM IAANKPGSII LVASMSGSIV NYPQEQSCYN ASKAGVIQLG KSLAAEWAKY  300
DIRVNCISPG YMDTAMNKVP SLDAQKKIWK SLTPQNRLGA VDDLNGLCVF LASDASGYMT  360
GSNVAIDGGY TLY                                                     373

SEQ ID NO: 9            moltype = AA  length = 361
FEATURE                 Location/Qualifiers
source                  1..361
                        mol_type = protein
                        organism = Kwoniella heveanensis
SEQUENCE: 9
MSVFRSTLST ARALSARPTF CAPSRAAGAG SIRAYSTPTK VVRKDEKGDP LEEHRVEVEP  60
KLAAIDETIT FEHPKKWGKH PGHDMTKGDF GRHTKRTLAS FSMDGKVCLV TGAARGLGNM  120
MARTFVESGA NAIVLVDLKK EDAERAAEDL VKWFVENGEA EEGEIQALGL GCDVSDEASV  180
ASVFATVKEK FGRLDAVVTA AGIVENYVAH EYPTQKIKKL LDINIMGTWY CALEAAKLMP  240
EGGSITLVGS MSGSIVNVPQ PQTPYNFSKA GVRHMARSLA VEWATKGIRV NCLAPGYTLT  300
NLTKVILDAN PVLRDEWLHR IPMGRLADPS DLKGAVIYLA SDSSKYTTGA EIVVDGGYTC  360
L                                                                  361

SEQ ID NO: 10           moltype = AA  length = 231
FEATURE                 Location/Qualifiers
source                  1..231
                        mol_type = protein
                        organism = Pichia kudriavzevii
SEQUENCE: 10
MEALHRTRDA MVRFCVEEAN IKEEDVPKME CFTCNIGDAG EVETLFGEIY NVFQRYPLHM  60
VNCAGYCENF AAVDYPAQNA HDLMGVNLLG AFYLSQCFAK PLIEHNISGG SIVLIASMSG  120
KIVNTPQNQC IYNASKAGVI HLAKSLAAEW GALMHPIRVN TLSPGYTATP LTRNVVSGDA  180
SLAAEWTRRV PLGRMAHPRE MAGAVLFLLA NDASSYTTGE DVLVDGGYSV W           231

SEQ ID NO: 11           moltype = AA  length = 282
FEATURE                 Location/Qualifiers
source                  1..282
                        mol_type = protein
                        organism = Candida maltosa
SEQUENCE: 11
MDASSYWSYE NIVPSFRLDG KLVILTGGSG GLAAVVSRAL LAKGADIALI DMNIERTEQA  60
ARDVLQWGEE QMKGKYESPI GQVSAWSCNI GDAESVELTF KAVNEHHGKI ASVLINTAGY  120
AENFPAEEYP AKNAENIMKV NGLGSFYVSQ SFARPLIENN LTGSIILIGS MSGTIVNDPQ  180
PQCMYNMSKA GVIHLARSLA CEWAKFSIRV NTLSPGYILT PLTRNVISGH TEMKNEWESK  240
IPMKRMAEPK EFVGSILYLA SDSASSYTTG HNLVVDGGYE CW                     282

SEQ ID NO: 12           moltype = DNA  length = 1536
FEATURE                 Location/Qualifiers
source                  1..1536
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 12
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ggggggcggg gtacgagtgt ggggagttca ctagttcgct ggctagctga ctagctgacc  180
ggcgtgtggt tgtggcagta cgcagtggga aagattaaat atgaaatatt gaatatcaaa  240
atgacgtttg ggtggcacgg tgcgcgcgcc aacaacaaca agagcagcca ccaagccaaa  300
gcaccaagca ccaaccccgc acaccatttt gttacacccg cggcagaaat ggcgtcagcc  360
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atcctgcagg agctcagacc ttgcatgtta gccttgacct tgcctctatg accactaatt  600
cgcacaagat tgcacgttac cattgtcatc ttgccatgta tccaatgttt cctgtttgcc  660
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cggccgttca agatcaagtt gtccctgaca atacacttgc atgggtctgg gtacgcggat  1440
tagatgaatt gtacgcggaa tggtctgaag ttgtatccac taactttcgt gacgcttcag  1500
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SEQ ID NO: 13           moltype = DNA  length = 4059
FEATURE                 Location/Qualifiers
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                        mol_type = other DNA
                        organism = synthetic construct
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gtacgcggaa tggtctgaag ttgtatccac taactttcgt gacgcttcag gtcctgctat  240
gactgagatt ggtgaacaac cctgggggcg tgagttcgca ctccgagatc cagctggaaa  300
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gaacacaagg ttacctacct ttatagttgg taaacagcaa gcaataccag agattttcat  3960
cgcaacgtga ctctatcatt cactatatta agttgcacac ggatgttgta caacactaca  4020
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SEQ ID NO: 14           moltype = DNA  length = 3810
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                         mol_type = other DNA
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gacatgtaat taaactctcg acagccatcg tccattgaca ccaacgtgca ccaacacacg  420
cgagagctgc ttcgttgtgc ctgtcgacga cattgcctcg ttccccagcc aagtcgcccc  480
accgctacac caagcgcact cgca                                          504

SEQ ID NO: 27          moltype = DNA  length = 493
FEATURE                Location/Qualifiers
source                 1..493
                       mol_type = genomic DNA
                       organism = Moniliella pollinis
SEQUENCE: 27
cctttgcctc acgcctctcg tttgcacgct tcttcacggc ttatcaccgc ttattatatc  60
catggaatcc cctccaaagt tttcttcaac atagcttcat atcttctacc ataacatcta  120
tcccaccaag tagcacatga attcaacgta gcttgagaac ttttcctcgt cagctgattg  180
cgttctttag gcagtttgct tctaccccaa ttgcttctac cccaattgct tctgcttatt  240
tcaccagtcc gctcggagat ctctgacaac tctgagatct gatgatcaac ctcttgtgca  300
acttgtaaaa tttgatcgat gagacgatta gggacttgaa gctttaagtt agacgcgatg  360
ggcggtaatc tatatgctta gactttgttg ctatgtttgt cctataagat ctgatgcatg  420
atgtatcatt acccagacgg gctatcgcac aaggtgcaga tgcatcttat catgacagtg  480
ccacactgtt ggc                                                      493

SEQ ID NO: 28          moltype = DNA  length = 499
FEATURE                Location/Qualifiers
source                 1..499
                       mol_type = genomic DNA
                       organism = Moniliella pollinis
SEQUENCE: 28
catccgaaat tggtggtttg acaaatgcac cttcttctcc aggggctcgt gcaaacgcag  60
tagccgagaa gaggcgtacc ggagaggaag acccaacaac cactgatcgg tttgaagttg  120
gatgagaaag ttgtgccaat cgcggagccg tagcaacgct acgagcgcta cctacaacga  180
ctccacgcgc aatcggtacg cgcaaaggaa cggatgagcg accgagcagg ttgcgcgaag  240
ccatgacgat tgctctaccc ccaacgagct ggatcaagtt gggcagaaca taatttatct  300
cgcttccccg cgccacaagt agctccacca ttcttctcaa aatagcccga ttccgctttc  360
gcggggcaaa acttccgcct ataccgatcc gattgaaacg ccaaaaaaat taagttgtag  420
tacgcctgtt ggtcgagccc gactggagca ccatttgaga cactaggaga taatttattc  480
ttttccttct tggcccca                                                 499

SEQ ID NO: 29          moltype = DNA  length = 501
FEATURE                Location/Qualifiers
source                 1..501
                       mol_type = genomic DNA
                       organism = Moniliella pollinis
SEQUENCE: 29
cctgaacgtc ctgaactctg tcttattaca taccattaat ttccctgcct tccctctcta  60
cgtcatcgga attttatctc ttcctgagca gcaaagattt ggtccttgtc cgggatacaa  120
cggagagtca tgattgggtt tatggaagag aggcgatagc acttgcattt tgaccattcg  180
gggattaagc gaacccccaa gtggcacaaa gagctcgtat gagggtcagt aagtatagag  240
ggacatgcaa aggagagaaa aggaaacaca atattgacaa atagagacaa gtacaggtga  300
tcgcaatagc atgatgtagg ggtatgcgca agtcacaatc gttcagggcg ggcataacat  360
cagcgtaccc gagcacaatg gttaatgatt aatgagcaga ggaaaaaata atgattatgc  420
cggggatggc aggactaact ggcgcagctc ttcagcttcg gcaacgcgtg cacatagtgg  480
tgtagcacca cgagattgag c                                             501

SEQ ID NO: 30          moltype = DNA  length = 300
FEATURE                Location/Qualifiers
source                 1..300
                       mol_type = other DNA
                       organism = Moniliella pollinis
SEQUENCE: 30
cgggtaagac caagtcggtt ggtgtttcga acttcagcac tcgtctggtc gacttggttg  60
aggaagcttc gggcgaacgt cctgcggtaa accagatcga agctcacccc ttgttgcaac  120
aagacgagtt ggttgctcac cataagagca agaacattgt catcactgct tacagtccct  180
tgggcaacaa tgtcgctggt aaaccacctc tgactgagaa ccccggtatt gtggatgctg  240
ctaagcgttt gaaccatact cctgctgctg tgctcattgc ttggggtatt caacgcgggt  300

SEQ ID NO: 31          moltype = DNA  length = 400
FEATURE                Location/Qualifiers
source                 1..400
                       mol_type = other DNA
                       organism = Moniliella pollinis
SEQUENCE: 31
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aaccagatcg aagctcaccc cttgttgcaa caagacgagt tggttgctca ccataagagc  60
aagaacattg tcatcactgc ttacagtccc ttgggcaaca atgtcgctgg taaaccacct  120
ctgactgaga accccggtat tgtggatgct gctaagcgtt tgaaccatac tcctgctgct  180
gtgctcattg cttggggtat tcaacgcggg tacagcgtct tggtcaagtc agttacaccc  240
tcacggatca agagtaactt tgaacagatc actctgtctg atgaggaatt ccaacgggtt  300
accaacctca tcaaggagta cggtgagagc cgtaacaacg ttcctttcaa ttacaagcct  360
tcgtggtcca ttgacgtctt tggtacccag gacgaggcta                        400

SEQ ID NO: 32          moltype = DNA  length = 500
FEATURE                Location/Qualifiers
source                 1..500
                       mol_type = genomic DNA
                       organism = Moniliella pollinis
SEQUENCE: 32
gcggcctatg cagacgcttg gagcttgggc acattggata ttggatattg gctggctgcc  60
gttcgagatc ccgttgctgt cgccaccgcc accggtcacg tggtcacttg gtcacgtgga  120
ccacctgcac tgcgcttagc tacaagagtg catgatgcat gatgtcatgc atgatatgca  180
tgatacaagc atgaatattg atttcaatat tgaacattta gtattgatat ctcgatatgc  240
gtaacacgtg acacgcacca tatcaatcag ccgcaccggc actaaatgtt tcaatttcaa  300
catcccacac gctcgaatgc tccctcgagc taatcggatt tcggagctcc gacatggccg  360
gcctcgagcg gagtttgaaa ctccgttccg cccgaccccg ctggtcgccg cgatgaaatc  420
ccgagtcaag tcatagcaag ccaagtggct ggctgactca aaagctaata aaagtcgtag  480
caagtaatac tccgcacgct                                              500

SEQ ID NO: 33          moltype = DNA  length = 500
FEATURE                Location/Qualifiers
source                 1..500
                       mol_type = genomic DNA
                       organism = Moniliella pollinis
SEQUENCE: 33
aaatcttcat ccagcgtagc aagcatatcg cgcatgtccg agttcctcga gtgtgcatac  60
cgaacccaaa aagccatgga tgtcattgac ctgctgtatt catcaatcga gatatcttta  120
ctccggagcg tacagtggca gatattatgt aggagatgat gggacaggat gcagagatac  180
ataatccatt tccagttaat agggccgcta gatacttgat attagcctgg cagcgagcga  240
gcgagggtag agtagaatag agtccagtat agatcccacc ccatggtgcg ggaagtggat  300
gacgtaatag cgcgatagct ccccgctccc ttgtcaccta ggccacggga ccctacgtga  360
ttgcatcgtg cgtgtcgtcg cctgtcgacg cctggtgcag gcatacgctg cgtgcttcga  420
cgatcaacaa caccaccaac cacgactcca gttccggcca gccagtacca tggcgtcgtt  480
ctttgggttt agtgcaccgg                                              500

SEQ ID NO: 34          moltype = DNA  length = 500
FEATURE                Location/Qualifiers
source                 1..500
                       mol_type = genomic DNA
                       organism = Moniliella pollinis
SEQUENCE: 34
tcgcgtcttc aaaccgtggc gcgcgacacg tgccgcgacc aaaatttgtt ggtgtatggg  60
aatagggttt tctaaaattc tgttgaggat gtgagtatca ctggcatggc accagtgtac  120
tgctcatcga taatcataaa cgtatcaacc cttcggctct gcgatctgta tctgctactg  180
gagggcacag caagccttct gcttggttgc gctctgcgct ctctgtggtg ctttgccgct  240
cgcggccaag gtccgcagcg ggcattcacc accatacggg ttaagccacc agacgagtgc  300
tgctcgccgg gccagctcgg cgcctgtcac gcccccacat tggcttgatg ctcggctcct  360
tctttcatct tggaaccggc gggcaatggc gcttatcacc actcctgggc cgggcccgc  420
gtatactctc ataactagtg cacgctttca aataaatacc tcgcctcctg gctggccgac  480
cgacgtttac ccatccgccc                                              500

SEQ ID NO: 35          moltype = DNA  length = 500
FEATURE                Location/Qualifiers
source                 1..500
                       mol_type = genomic DNA
                       organism = Moniliella pollinis
SEQUENCE: 35
atcctgtgct cgtctcttac attcatcttc ctgcccataa aactttcaag ctatttcctc  60
caacgtactc aacaacctat ttttttgggc cccgctcaaa acgcacctta ttgcgtctaa  120
gtgtattcta accacgcgac ctgaattgca aagcctgtta cccccatcgc accttcgtct  180
taagatagtt gtttaacgag ccttggccag ttttagcctt gaccatgact acagatcata  240
atggttagat gcttcttgat cgataatgat atattgcatg tcgacgattg aagatttatt  300
ggatgtcgat gacgactgtc gagctcttat tcttagcaac aacttaaata ttcactatcg  360
aatgcacgcg acatcatgca tcgtgctcat gaatatcact ggatcactct cgatcgcagc  420
ccactaggtc agcataacac gcttgaagct gacttgaagc tttggaacaa atggcagcaa  480
cagtggcggt cacggtgacc                                              500

SEQ ID NO: 36          moltype = DNA  length = 500
FEATURE                Location/Qualifiers
source                 1..500
                       mol_type = genomic DNA
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                         organism = Moniliella pollinis
SEQUENCE: 36
tgcagagcct tgcaaatgat gaaagaccaa tggaattgat cattgattat tgatgatgca  60
tgatgaaaga gatagaaaca gatgcaagat gggatggcgc gatgatcaaa tgataatgaa  120
cgttcaaccc ttcaaaattg aaactctcag gctgagcggc tgagcaagct ctgagaatgt  180
gggtcttcga agactttcgc ttgagcatgg tgcatgtgca aagagatgga atgtggaatt  240
gttgtaggta ggaacacagc ccggtagagc ggcgcagtgg ctcagcgaag ctgggagaac  300
gactgaaccg gtcttgagat tcatcatctc agcgtgcgtc agcgcgtcag cttcagcgtt  360
tgagaatggc acggactgac ctgtcgtaca tcccaaggag gcccgtgcgc tatactcgcc  420
ttgtggccta atctgcgatc aacccatcct caacgaaagc gcaagtagcg actaagcaca  480
ctatctttcg ggtcggtgcg                                              500

SEQ ID NO: 37            moltype = DNA  length = 486
FEATURE                  Location/Qualifiers
source                   1..486
                         mol_type = genomic DNA
                         organism = Moniliella pollinis
SEQUENCE: 37
attgttgcat cgtctcatca gatcgtcctt cattgtagtg ataattggtg caaccttgta  60
ttagactgta tagcagaatc cctcgctata aaaagtatta tgcatacgca gttccaagca  120
ggaattagta tggtagtggg acttggctga agggctgacg acatggcgag aaaaacgtat  180
caattcggaa cacactgtgc aatgtcgatc acgttttcgg tggttcacgg taggcttcct  240
agtccatggg aaagtggtaa catctatccc ttgagcgctt gagcttaatg ttaccttgtg  300
caaaagtcat ttgaagcttc acttgaagct tgaagctttg aagttttgaa gcttgaagcg  360
caaacagcag cggtacatgt atactggcgt aacctgccta cagtggtgca tcagtacgtc  420
aagcgggtag cccgaaatct cattaaggtc atttgccctc taagcttagt acctagtgat  480
ctgttc                                                             486

SEQ ID NO: 38            moltype = DNA  length = 499
FEATURE                  Location/Qualifiers
source                   1..499
                         mol_type = genomic DNA
                         organism = Moniliella pollinis
SEQUENCE: 38
tggcatgtag gcctgtaggg gctatgccta tatgctcatg cactcatgtc gcgatgttgc  60
gatatcgcga tggaccaacc aagcaagcgt gcagcgtggg cgcttgggcc aagggtgcta  120
agctaagtgt aagcactgaa gtgctgaagt gctgaagtgc tgaacgtgga acgtcaggat  180
gacacatcct ctcccttcgg cttcgtccaa gaaaggttgg tcagccatct gagatctgag  240
atctgaaatg aatataatga tgaattgagt cgatgagatc tgactatctt gggagattgc  300
accaatgcgc cacggtgacc agatggtctt ggtcttagct ggtatagtgg cacctgatct  360
gccttgacag atctcatgtt catacgttca tgcattcaat cttatcgtct agccgtctgc  420
gatctgccca actagacgac gacaaatgta attgatcctt gtaaccctgt tgctggtcgc  480
agcagcgcga tacgatgcc                                               499

SEQ ID NO: 39            moltype = DNA  length = 504
FEATURE                  Location/Qualifiers
source                   1..504
                         mol_type = genomic DNA
                         organism = Moniliella pollinis
SEQUENCE: 39
ttgctgctac gacctcattg atcgatgatc tccgacactt tgagtagctt tccatcctct  60
ctttactgta ctttgtattc ttgctacttt gtgttctttc ttaatggtcc cgaaccggtg  120
attttgctca tgatcttttt tttttttttt cgtcaactgc aagaagatga aaaatctact  180
gttcgatagc acctcataac cgttccagag tcccgacaac cacaaggatg tttcttagat  240
aggtgtcagc aagatccaca aaaaagtgta ggtattagat ggtttcaagc aaggtacaat  300
attctaattc aatactcatt tgaataatca atattcaaca gcttctagcc tacaagcttc  360
aatgttctcc gtcggttaac tcagtatact tagtactcag ttcaagtcag tgttgttaaa  420
ataaaatcac gtgatccaaa aacttgaaac ttttcatttt tctcctcaga tgttggtgat  480
tcgtcccata gcggttacca caac                                         504

SEQ ID NO: 40            moltype = DNA  length = 1002
FEATURE                  Location/Qualifiers
source                   1..1002
                         mol_type = other DNA
                         organism = Moniliella pollinis
SEQUENCE: 40
aacacgtaga acacagggat cgccatttgt ccgaaatgca atctttgctc tcgaagcttg  60
aagcttgaat agcttgaagc atgagtgcat gaggggctgc tatgatcttc caaccaaact  120
tgcttgggca ttgcactttg gctggatagc tatcatttag ctgaccccca aagagactgc  180
caatgatatg atatttgtgt aggtgcacga gtggatgagt ggcgcgacta cctgtcacaa  240
gatagatgct cgatgcactt catgtgaagt ggctagtgga tgtccggaga ctggagtaca  300
gtaaagagtt tggagttggg aacagcgcga atgaaagagg ctcagagaga ggtaggtttg  360
atgtgacgtc aagcgggcta gtcagttagc aagggagtat ggaatgcagc gcgcaacgcg  420
cagcgcggag cccagtgcgc ctgcgctcag gtactcatac tcagtactca gtactcaggc  480
gccatgccgg agcttacatg caatcaatta accctgacgt ctggcgcggc ctgacgcctg  540
aggcctgagg cctgacgcgg ccaagctcca agcgcaaccc caggctggct tgcgctggcg  600
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ggcgcgagac gcagcagggg gcaagtgtcg cagagtcgcc gatcggatct gtggaacagc  660
acaggaggcc cgtgcggtcg gctgcgcagt gcggtggaga agctggaagc tggaagctgg  720
aaggtggaac ctggaacttg gaacttggaa aggaagaaga gcgggaaggc gatgcacgat  780
gcacgatgca cgatggggtt ggattgcgta cgagtagccg cacagcgcac agcgcacaac  840
actcacccca tgcccccacg ccccagcccg tgctcgggcc atgagctgct ggcaaccccc  900
cgaggcagcc ggcctttaac tcaagtagca ttcgccgccg gtcggttctt cttagcctcc  960
caccaccacc acttgcccgt gcttggacgc taattacaat aa                     1002

SEQ ID NO: 41          moltype = DNA  length = 679
FEATURE                Location/Qualifiers
source                 1..679
                       mol_type = other DNA
                       organism = Moniliella pollinis
SEQUENCE: 41
agctcagacc ttgcatgtta gccttgacct tgcctctatg accactaatt cgcacaagat  60
tgcacgttac cattgtcatc ttgccatgta tccaatgttt cctgtttgcc atagtacgta  120
gtagatacat cgatcatcga tgcagatgac gcagatggca ttggttcgct ttgccctatg  180
catccatcat gcatcatgca tcatgcatca tgcgtcacta tccatattaa tcattgccct  240
ctgtactagg ccatgacacc tgcagaccct tccccaggga tgcttcttgt aatagccaat  300
atactacaca actttaccga ttacgatcga gatcgatctc actatagttt gtcgaacttc  360
gatgcacatc gagggtgcgg aaaaaagctg acggaagcgg aagtgggtga attggttaat  420
ttagttagat tggcagggtt gttcccaaga ttcaacgcaa gctgagaaag aaaacagaaa  480
cacgaggcat aaggccagtc agaaaatgca ttacgagggg agttgaagtg actgtgggcc  540
agtgcctggg tcgtcatgtg caaatttttt cttcggttag actgccgcag catgcagcaa  600
ctctgccggc cctatataac tagtgcttca tcttcttgtc attcttgtct ctttaccaac  660
acaccggtac tctttcacc                                               679

SEQ ID NO: 42          moltype = DNA  length = 1002
FEATURE                Location/Qualifiers
source                 1..1002
                       mol_type = other DNA
                       organism = Moniliella pollinis
SEQUENCE: 42
catcgttcca gcttgccttg ggtgctagcc agcgcaaaag tccgacgaac agacgaattg  60
accgtagtcg atgacagtcg atgacagtcg atgacagtcg atgacagccg acgaccgacg  120
aacaacaacc aaccgcacgc agcatgaggc cgagaggctc aggctgagcc gctaaggcgc  180
tgaagcttga agctcggtgg tccagtcaga tctcatgatc tcacaccgcc gcgctgctgc  240
ggtcgtatgg ttcaatttgc tgtgcctagc ctagcctggg tacgtcggta actagccact  300
agccactggc cattatttgg ctactgggca cactaggagg cgcactagtc catcgcaagc  360
ggtaacttac ctaccggcca ctacctgatg atttccctct tgcctggcgg cctggcggcc  420
tggcggcgca ctctttgcgc tcggacccgc aaattgcaaa ccaacaaacg actagtgtcc  480
ggtacgtgtg cccaagtaac ctgggcagac atagaaccac attgatcgta gggaaccgca  540
atggaccagt acgctattct cagtacattc tgtacacttt tgcctagatt ggaatctggt  600
gaggatcaag atcaaatttc aaaccaactc cacggcactg gttcccgaca cttggactgt  660
agtactgagg tactgaggta ctgaggtact gaggctcaac tgagtctcaa ctgaccgtga  720
ccgcgcttgt gtggtggcgt ggattctagt gttggatgtg cgcagtggtg tgtgcggtgg  780
tgcgttcaga ttcgtcggta gaacttgact attcacgtct tgtgctgcgc tgacgcttgg  840
aacctggaat gttaaacctt gggtcctgag ggtctccacc ccccaagtcg actgggttct  900
atataaagtc gggtaggcaa ttggctaggg agaatctctc ttttctcttt ccccagaaca  960
cctgttcact cacttctcac ctctttactt acctacttca ca                     1002

SEQ ID NO: 43          moltype = DNA  length = 1000
FEATURE                Location/Qualifiers
source                 1..1000
                       mol_type = other DNA
                       organism = Moniliella pollinis
SEQUENCE: 43
cctcgttctt gattgctttc aaaagtaatt taaggagaga acctggtcgt ttgttgctct  60
actgaatgat cttacctata ccagcatacg acaccttgct gcccctcccc ttcaccacca  120
ccatcaccac tttcctcttt ttctttgggt ttttctagtc gttgcattta tggttgattc  180
gctctttgag aaccgtatgg ctgttattcc ttcccaactc ttttgtctgt tttatcgttg  240
catccaaacg tttgggctct tcgtacctct tttttctctg tttgcctttt atttttggtt  300
tttctatccg tttgatggtt gtgactcttg ttttcctttg ggaaggtttg agctcaactt  360
ctcttcggtc cttgttggag tttgacttcc ttttcttttt cagctgtggc cgatcattac  420
gatcaataat ttgtgaatat gatatcgact acatcctgaa tcttttagcg aaccatagca  480
gtttagcgtc atagatatgc atgcatgatg tacatcttac acgcaccacg gatgatgctc  540
agcgctcagc cacgcagccg ctcagccgct cgctcggtac gctcggtcat gatgtgcatg  600
atgcaagcct aatgattgac gggtacttga tctgtgtccc aattcgatcg agatcgatct  660
cactatagtt tgtcgaactt cgatgcacat cgagggtgcg gaaaaaagct gacggcagcg  720
gaagtgggtg aattggttaa tttagttaga ttggcagggt tgttcccaag attcaacgca  780
agctgagaaa gaaaacagaa acacgaggca taaggccagt cagaaaatgc attacgaggg  840
gagttgaagt gactgtgggc cagtgcctgg gtcgtcatgt gcaaattttt tcttcggtta  900
gactgccgca gcatgcagca accctgccgg ccctatataa ctagtgcttc atcttcttgt  960
cattcttgtc tctttaccaa cacaccggta ctctttcacc                        1000

SEQ ID NO: 44          moltype = DNA  length = 1007
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FEATURE                 Location/Qualifiers
source                  1..1007
                        mol_type = other DNA
                        organism = Moniliella pollinis
SEQUENCE: 44
cttgacgggt tcaatattca atactcagtg ttcaagcagt ggtaggttgc tccgtggatg  60
gaacgacggg tttaatttct tccgccatgc accatggtcc atgatcggtt cagatggaca  120
gttggtatga gaaagtggta acagataaat agaatttcgt acaagttggc aagagcctgg  180
ccgatggtcc atggctcgat ggctcgatgg ctcgatggct cgatggccac acaccatgtc  240
atctactttc tgtgcttcta tgcggacgag tcggaacagc tgtcaaactg cgtttggggt  300
atttgagcgg aatttgaaac cttgttgaaa ctacattgtc tcctcgttac ttctctggca  360
tctgtttgat cgaacggcgt catgcaatag acacgcgatg aggcccaacg acggtgacgt  420
tgaacgcctg gcccacctgc tacaacaccg tagaacgtgt tgaagttgaa cgtcggacgt  480
tggacgttga gttgacgcgt gatgctggaa ccgtggatgt agcagcgaca gacaagccct  540
ctaagcaatc tagtcgattt tcaaaccaac taaacgagcc ttagttcgga caaccctagt  600
ccagtgcggg ggcagcgtgg ggtcagttgc cagcgaaggt caccccagct agctgggttg  660
gctagcttga ctggctcaga agtatcgatg ctggagcttc aggccagtac gaaaatacac  720
tgtcgtagtg ctccttccgg agttctgggc ttctgagccc ggcggttgtg catcggtgca  780
cagtgcacgg accggcgggt gggacaacaa actgtctggg cgtgaaatcg tgcgagtgtt  840
gatgcaaacc ggacgtaccc agcacgcagt tcccagtact cagacatact cagtacctca  900
gtggttgagc ctcagcatcg cctccgtatt tctacaaagc tgttgagtcg ccttttgact  960
ttttactacc accatcccct cttcttctaa accccttaca ttccaaa                1007

SEQ ID NO: 45           moltype = DNA  length = 1000
FEATURE                 Location/Qualifiers
source                  1..1000
                        mol_type = other DNA
                        organism = Moniliella pollinis
SEQUENCE: 45
tcgacgacag tgagtatagt caactatgct atctagtagc caaagtaagt ctacattccg  60
ggctttgcag gggttagcta tttcattaga caatacttac aaccttcgaa gggcttcccc  120
agaattcaat tctatattga ttcaatatcc acccatcgga cccgctctcc gtaccccctg  180
ttcgcttcgc cactacaacc atgtcaagac ctgagaatcg tagcaaattg ccccactacg  240
gcccgagtgc tgtcggctgg ttaaagtaga atgacttggg tagccagatc gagacgcctg  300
acattgatac gcagacagcg cagtcagttt caaactgaca actgacgcgt actggattga  360
caggactcct tggccggtgc cattcgaatc gcttgtagac ctcttgccag catcttgcca  420
gagaagcttg aaggctggat cgctcagcct gagctcacac ttagtagctc cactagtcac  480
tcaggtgctc agcctctcct cagcgcctca agctcaagcc tcagggcctc agctgcatcg  540
agctctggtg ccagtgccag tgccagtgcc agtgccagtg ccagtgccag tgccagtgaa  600
gagagtggta tgagccaagc tatacccaaa agcaaatgat ggttattgcc agcttccatg  660
cacccctggg gcgaaggaca acgaacaaca gacagagttt gaattattca ctggcctagt  720
actaccaatt taaatttggt ggttcagcgc aggaacggcg cgagtgagcg taagggcaaa  780
ctccgcatgg ggcgtgacgt cagtgtccct ctggctgagg cactccagca ggctctggtc  840
aacgcatcaa cgtgtgtgtg tgtctaccaa gcacattgtg cgctcgctgt ccgtgcgccg  900
aaatcgcacc agtcagaacg tctataaaac ccctcttgct tggagggtta ccgggccgtc  960
ttttcccaac gctccggctc actcactact ctacacaaca                        1000

SEQ ID NO: 46           moltype = DNA  length = 997
FEATURE                 Location/Qualifiers
source                  1..997
                        mol_type = other DNA
                        organism = Moniliella pollinis
SEQUENCE: 46
tcggacccca agctctctag cttcaagctt tcaagctagc ttccagccag ggtcatggaa  60
ccgtaggagc tgggaaacgt gacgttgtcg ctatgcctat cccatataag aattaacgtc  120
gaatcatgag gcgcatggga ggctgcattg gtggtatgat acataacgtt atgattctat  180
atgagcgatg gagcgcggcc gtctgccgct agtgtggaag ttcgatacgc ccagtccatt  240
tcgttcagat gacgtttggt ccagaagtac ggtacatacc gagcgcaaac gtcgaactta  300
tggccttggt cgctcgtggt ccggcgtcgt cgacactttc atcaacggac gaaaggtaga  360
gccgaacgcc agcggccaat ggcgatggcg atagtagaat gaatagtcgt cagatctagg  420
tcttaggcta caggccaccc tgcctgaggc ctgggggtct tacagcgagt ctaaaaaacg  480
aaaagccaac gtgccaactt gggcctgccc atgcaccctg ccagagagcc ccaagctggt  540
ctgatgcaag cggagcaaaa tgttggaaga atagtttcaa cgatggtttg gttccagtac  600
gcacagtcga gagatgaggt actgcgcaac gagttgttgg cgacggaacc gaagatggat  660
tgaactattt gatctacagg tgatggcgtg gagtcgggcg tggggccgct gtgtgatatg  720
tagaagtggc gctattcgag ggctgcgccc gtctggatgg atgcagaatg aagcttgcag  780
gccgcagctg gcaggatggg cgcagaaccc ggtgaaacat agttaaagcg cggggccagt  840
gtcatatttc gcttgagcca ttctactcgt tggcagccca ggcgccgcta atttacgcca  900
actcttggta ctcgagagct cgggcaatgt ggtatataag cttgaggttc cctacagcaa  960
agcctggctt cttcaccaca accgcactct agcaaac                           997

SEQ ID NO: 47           moltype = DNA  length = 1002
FEATURE                 Location/Qualifiers
source                  1..1002
                        mol_type = other DNA
                        organism = Moniliella pollinis
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SEQUENCE: 47
gagcgtagag agctccatcg atggcctggt tacaagttgc ctgaagacca ccaatacggc  60
ccgcacgcca acggcaggcc aagaaaggat aattgacaac cgtgaatctg aaagcttcgg  120
cgctcaggag ctcatgtgtt ttgttgtaga tataaaggtt tctgaatacc agttgggtat  180
tgacacgggg agttggactg aatctctcag atctcccaga tctcaagtac tgagtcccaa  240
gcgtgccata attaactcta ggttgcgcgg gacacgccac ctgagatcag ggtgtacgcg  300
catactaatc cattccacca ccaaagcaag actgagggta ccgcctgagg ttctggctat  360
tccaacagtc acaagtcacc gcgattgccg tcaaattcca aagcgtagta ggcgctcgac  420
ggagggatga tggccagtac ccaatttcga gcactcagtt agatctgttc gtgtttagtt  480
gctcctgcag atctatctta aatgtatact gtgccaggta ggatctctcg gcacctatag  540
atgatataaa agatatgatg atatgccaag tgccaagccg tcaaactgac aaaatgaaag  600
aaacagcagg caagctaggc aagcgtctat cgggggatta aacggatgag tcagcagcac  660
aggagcagtc agtcgacaat tccggctttt tcccgcccgc aaggccccag accccagacc  720
ccagccgcca ccatgtccgt atccgccggg gttagcatga ccacggcaga tttctccgct  780
cagcttgctc ggacgactat atataaagaa tcctttcctc gcagtccgtc cgagcccttc  840
ctccacccca tcggttacca ccaacgacat cacttggtct tgacgccgcg ctgcgttttc  900
ccgcttccgc gtctccctag tgatttagct ggtgttttcc tctacccgcc actctactgt  960
ttcatcgctt gttgccaagt tttctcagtt cattctcatc aa                   1002

SEQ ID NO: 48          moltype = DNA  length = 1002
FEATURE                Location/Qualifiers
source                 1..1002
                       mol_type = other DNA
                       organism = Moniliella pollinis
SEQUENCE: 48
tggacgtgga caagcaccct gaagtcgctc agggcttcca gatcaaggct ttgcctaccg  60
tctacggctt gtatgagggc aagcctgttc gtcagtgtaa gttttaaaat atattgttct  120
tgtctgttcg ggttttacat cagtggtgta ctgacacggc tttttgtttt attgctgcag  180
tccaaggtgt acttcctgag gatatcctgc gccaatttgt ccaagatctt cccagcttgg  240
ccgcccaaga tgcccagaac ccgcctgctg aaagtagcga ggctgccaac taagtatagt  300
ccatcctatg agctgtccta ccccaaacaa ctttttcacg cccaacgtcc aaacgtcact  360
tcttattgcg gcaaagaaca gctgcttccc cactcattca ttaccccttt ctgatccacg  420
aggtgtaccc atggcctttg tacattccat gaaacaggta tccttttccc caccattaca  480
tctgaactta caaagacttt gcttcctctt gtacacagct cgtcgtccat cgtccatcat  540
cgatgacgat catcctgcat tcggtatcat gcaccatcat gcagcgatgg caacgagggc  600
catgcaccat gctgcatata ctatgtgtat agccaatagc caatcttcaa tctccaactt  660
cctacagaca cagcctgtgg ggaatcgctt aatcgctgtc catgcgtcca tgctccgcgt  720
tcctcgcgct cgtcgcggcc tcgcgacgat taagtcacag tcacgcccac acagcgcgcg  780
agcgcacgag ccaccatcat cacgtgatga gacaaagagt ccacttccgt aaaagcaatc  840
ttaataatac atgtaaccct agcaatgtag ccctaataat ttgtttggcc tttattttat  900
tgcgccccccg cagcttttgc accccggaga gtaaaagtat tccggttgct cgtcagagca  960
tccctttctt tctccatcag caaagcgtgc gaccgccaag aa                   1002

SEQ ID NO: 49          moltype = DNA  length = 1002
FEATURE                Location/Qualifiers
source                 1..1002
                       mol_type = other DNA
                       organism = Moniliella pollinis
SEQUENCE: 49
tttagctacg atggtgtgcg tgtaactgat cagtacaccc gctgtatcaa aactctatct  60
tcaatccgct aagaatcgtc accatcagta catcagtaca tcagtgggct aagatctaat  120
ggtgtcgttc gacggccgcg ggcactagcc acgagccaag agcggtacag ttgctgcgaa  180
atggtgggca gcctgacagg ttcagtttga ataggatgaa aacgccatga agaacagaga  240
agctggcaat tgacgatagt ggggcgagcg cacgctgctt gctggtccga gcgtaggacg  300
aacgacaaac catgggcggg gccaccaggc cccagggagg gcgtcactgg ttggatgtcg  360
tctgttaggc tagagctcag gcgctcaggg cctctggccc aagcctcgga cgcaagcctc  420
tgacgagcga gcgcgcgctg gggcgggtct gcgcaggaac ggcaggactg ccgagttggt  480
tgaatattga atggttggat gattggtttg ctgggtttgg ttttgacacg cacggcacag  540
gacaagtgag tttggaggtt ggatttgcgc aggtagaaag ggcgctcggc ctcagggcct  600
cagcgtcagc agacgagctc gccagtccag gtcagtaccc agtgttcgtt acttcccacc  660
gtccagagtt tgctgctgcc tctcagcttg attcgtgtcg gagaagtaga aatggtcaca  720
gcgtaccaca cactgtcccc atgcctgcgt cctgctccac gagtgtttag ccacaaaaat  780
ttatggacaa ccttactgag cctgtgctgg ggggtgcttt agcaccagca ctggcgccat  840
ggcgtcaagg ttaattccac tgtcattcga caaccttcaa ctgatggggg caactgggca  900
caaaaccttg gatatataac caactccaac tcccacggtt ttttaccttt tgctcttctc  960
cccaggactc actacttcta ctcctctatt catcttcacc aa                   1002

SEQ ID NO: 50          moltype = DNA  length = 921
FEATURE                Location/Qualifiers
source                 1..921
                       mol_type = other DNA
                       organism = synthetic construct
SEQUENCE: 50
tttggatgag gcattcaatg tctccatgtt tgaagaccag agaaagtgta ggtaacacaa  60
tagccatgga caattagcga atccgaatat gtgtactaac attgtgcctc caactgacag  120
ctggtcgaat aatgattgaa atcatcaaac atatcagaga tatatcccag aaccatttaa  180
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caactgttat tgtatgtatg cattactgtt ttacattgtg tctttccaaa tactaacatt  240
gccattatcc aatccataca tatatatata tatagattac acaacgcaaa cctcgacagt  300
cgattggttt acatcgtgta agcagccttt tacggaatat gatggacaac cagatcctca  360
tagatggaga caatgtgcag atagattcca tttatgatga cattgatgtg tttttaaatg  420
caaaataatc tcctatataa gtgagttccg agtctcccca tgcttatctt atcgtgtttt  480
ttctgtaaat taggtcaaag tattcaaaca attccctcaa tctttcccat tttttgctga  540
gcccccatat catcaaatta acgtatcaaa tttcctaatt agtagccccg tatgttaaat  600
tagacatgtg tgactgccat cccggacagc ctacccaatg acacccgcgc accgcacata  660
ttgtgtttag tgcgcgccgt ctgctgaagc gactccctgt ttgggaggaa ccgagggcgg  720
gttgcccgat cccttgcccc tcgctcctcc tcctgggctc ccccttgcag agggacaccg  780
aggggatccc tcgtgtgaga gcttggaggt ggatggtggt caatttttctc atttgattga  840
aagatttgtt atattgaaaa ttcagtttgt ggaagttgtg attaaaaggt tttactgttt  900
gttctgtaga cacattcaat a                                            921

SEQ ID NO: 51          moltype = DNA  length = 500
FEATURE                Location/Qualifiers
source                 1..500
                       mol_type = other DNA
                       organism = Moniliella pollinis
SEQUENCE: 51
acggatcaac aaaaatgtgc gccagctttg ttgtgtgccc ttttcctcta gcggatccta  60
tccacatttc gtgaatgaac cgattaagct ctgacgagat ttccttgttt tctgtagata  120
cctcttgtag aggtgtcaaa gttatgttgt gatcgagaga gttacctgca aggtattatg  180
caagcggctt gatgaggaca ctcaactacg tgtgagtagg agtgcgttgc tcagcagttg  240
acgcgtgagc cgcttgagca gcttgataaa ggtcctaatc gtcttccagc tgttggcaga  300
cgacaatgtg ctgtagtagt aggataggcc cgtaaaagtg tagaagcaat aggttgagac  360
tgactgtaat gggtacttgc gttcaaagac tgaaagtcca aaggattgct caaaggcccc  420
tgaggtctga gcctctgagc gcgctcttcg tcaaaattcg ccggggccat cggtcatcct  480
ctcgagggca gcatcttcct                                              500

SEQ ID NO: 52          moltype = DNA  length = 500
FEATURE                Location/Qualifiers
source                 1..500
                       mol_type = other DNA
                       organism = Moniliella pollinis
SEQUENCE: 52
gtactaacca agtctattgt aaagctggtg tctgctgctg cttcagcaag tgccattgct  60
accatcgcaa gcaaattgca aaaaaattcg ttcaatggtc taatttaagt aaaatactca  120
taatttataa aatgccgctt gtccttacct tttgtctttc tccttattat agtagctgta  180
gttatcgtag tcctcaggac ttcgtcaacg ctcgttgatg atcatatata cttcacctca  240
ttcattcata aagcattcat taggattgta tgatccggca gtgcatcatg catcatgcat  300
catgcatcag gacatcatgc atcatgccgt atcggtcgat cattgcattt gattggataa  360
tacgatcgtg tttaggcatt ggcagtcgaa acgaggataa agaagaggtc agtaccagtg  420
cacagcacgc gctcgcgctc tatgcgcact ttgacgcttg aaaaaaactt ttcagccaat  480
cggatataga cagcggaccc                                              500

SEQ ID NO: 53          moltype = DNA  length = 493
FEATURE                Location/Qualifiers
source                 1..493
                       mol_type = other DNA
                       organism = Moniliella pollinis
SEQUENCE: 53
atggtctgag ccgcccacga ttccttcctt catgagcgta tgacgaccct tttacttagc  60
atgtgcctct aaagaatacg ttttgcatac ccacatattc gtaaccctgt cctctatagc  120
ttctacaggt catagtcact ggctgccagt ataaacgtgt caatatctaa tcctcataag  180
atgcctcgct ctgtcgtcca caataaaatt cagtaacgca aggactgcgc gactgagtga  240
ttgccgatga tatgagaaaa ctaatttcaa gccatttgaa cttcaataaa tgcttccgct  300
aacatcacgg atcatcgatc atcaaccagc ctccattaat aggcctgcca ccactcgcgc  360
aatcatgaat ctgatcatac ctggctagcg attctcataa caccagaatt acctgcccat  420
taaagaaacc ctattctgcc tgtaacccgg caccagccgg tccgaattca cgacagacta  480
aggctctcgc gca                                                     493

SEQ ID NO: 54          moltype = DNA  length = 500
FEATURE                Location/Qualifiers
source                 1..500
                       mol_type = other DNA
                       organism = Moniliella pollinis
SEQUENCE: 54
atcagttgga atcatcaacc tttttttttt gtatcgattt cttcctaccc tcacctccct  60
tttccccctt aactcctcgc tttgaacctc cttacccgag tgcctacctt tacctttttc  120
tttttctctc tcccctgcct aatattctag gcttgctact atactgctct ctctaacacg  180
atttttgcgc ttgtaagcat ttttctcttt ttaaatctag gttttatcat ttgttgcatt  240
tatccaagtt tttttttttt tttgtttgta gtggttttac tgtttttgtg ttttgtagct  300
tgactaggtt cttattctat tgtagatgta tatgatcgga tgcattcccg atctaaaagg  360
aaaatgttct cgactgatat ttatttttac cttattgtct tcctaaatgg tttgactctt  420
gccttcttcc ttgggacgct ttgcaacgtt ttgtacgtac gtaagggttc atcgtttgga  480
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taagcgacac agttgtcaag                                          500

SEQ ID NO: 55          moltype = DNA  length = 324
FEATURE                Location/Qualifiers
source                 1..324
                       mol_type = other DNA
                       organism = Moniliella pollinis
SEQUENCE: 55
attcaccctg gagtaactgc gcttcatcgt ttctttgtta cgacaataat accacgctgc  60
atgctttgaa ttgagtatat attttgatga ctggttcaaa tagatctttc ctccatagcc  120
ttcgagtact cgtgatctca gcattatcag atttgtgtcc aggcccccga ggacgatgcg  180
gttttcctgg agtcactagt tgggaagatt tgtactggtt gggtgtaggt agagagaacc  240
attgattttg gtgtagctgt tcgtttttata aggccgagat tacgtaatgc acatgctgcc  300
ctagcaaatc gcgtcgcgtc acgc                                     324

SEQ ID NO: 56          moltype = DNA  length = 496
FEATURE                Location/Qualifiers
source                 1..496
                       mol_type = other DNA
                       organism = Moniliella pollinis
SEQUENCE: 56
atgtgcttag attaacatac acttgacttc cttacgcgtt gatatatact tggatctcca  60
taacttctat tgtatagttt aatcaatgga ctccatgttg acatcccatt atgcgatcta  120
tagtagctgg aaatagcaaa attgaagttc ggcagacgac tgcaccagga ggggtagatt  180
ctgtcccata actccctacc acaggccgtg ctctggttcg atttacctta gcctgatgcg  240
ctcaatgtaa cgcgtaattc tgagtggggt tcggtcgtcc aaggtttcag tatgcgctgt  300
gcagatagta ctgtatcgtg aaactcgacc gttgcacatg cactttatca agtgtcgtag  360
gcccagagtc gtaggccaag agtcgtaggc caagatggaa ggcctgaaga aaaagaaagc  420
tttgatcgtt ctgatctatg caatggattc atgagctaag tattagctat acaaatgatt  480
cgagcatagc agaagg                                              496

SEQ ID NO: 57          moltype = DNA  length = 500
FEATURE                Location/Qualifiers
source                 1..500
                       mol_type = other DNA
                       organism = Moniliella pollinis
SEQUENCE: 57
gttgctccct ttgctgttct atgatccaaa agatgcttag tgctccactc tctactcccg  60
actacctttt acgcattctg acatgactag tagtgcaatg ttacccctgg tgttttctcg  120
caatcacttt ctgtctgcta tagcttttga tcgtcagtag ttttttggta tagctaaagt  180
agacttttgt acctggcgtt caatgtaatt ttactccttt tttgtgcgat gcaacagatg  240
gtgtaaagat cgtcgttaaa gttttggctt gatgcttgat tcggtgacgg ttgcttgggt  300
gatccttata tcgtctcctg gtgttctcat aagcccaggc tcttcttagc tatgtacaac  360
aggtccacca ctgactatgt tatgtagagg catgcattta acttacggaa agatttaatg  420
gagaagaacg tacctagtca gatgcttgac atgaaaacgt attcctgcaa ttcacaaatc  480
agtatcctgc ctgtccactc                                          500

SEQ ID NO: 58          moltype = DNA  length = 500
FEATURE                Location/Qualifiers
source                 1..500
                       mol_type = other DNA
                       organism = Moniliella pollinis
SEQUENCE: 58
gcggttcgca cttttatccc aggcatgaaa aatgacgcac ggctgtcgca cgctacaccc  60
agtagtcgcc ttttcaggca gtctatgagg ggagcttttc attgcccaaa attactagcc  120
tgcaattcat gatatcaggt tcactttctc ccgatgtcta tagatttctc atagcgcttt  180
gaaaagacaa tttagctact cagtttataa gcaaatgatg tggaggtatt tattaggtac  240
aagatacgat aacgtttaca cgcggggaag caagtggcaa atcgtcgaca acaccgcatg  300
gtggtggtgg aatcgtactc catcaaagct tgcttgtcac tttgggaggc accaggttga  360
ggccatcgcc cgaaggtggt tgtggccagg gtttcgatat ttcatgaatc aacgtgttga  420
tggtattgga ttgggtaaga ggtcgatccg gctgttggtc agagtgctca tcacattcgt  480
cgttgtcaag cccgagttgc                                          500

SEQ ID NO: 59          moltype = DNA  length = 500
FEATURE                Location/Qualifiers
source                 1..500
                       mol_type = other DNA
                       organism = Moniliella pollinis
SEQUENCE: 59
gtggcgttga ttcgtatctt attattgatc tgtcctgttc cccacgttgc caagtgattg  60
ttcatgctat cctctctcca ttctgtttgt gttgcacagt cggtagttgt acgtagtagc  120
tcgtcacaaa atatgaagtg agccttgcaa tgtaggcatt cacaagctgc taggcaggct  180
atgatctaag ctaagtggca aaaattctct tgattcaatg ccaaactata ctattgttcc  240
tcaggcaaaa gcagctatat aagttgcatt caacgcaacc gccatacaag caaattagag  300
agaaacggga tataaggtgc tttatccctg aaggagtagc cagcgtctac ttggttgcta  360
cacaatacag caaacagtgt acttaatgag ccattggctg aactggagct cgcagaggtt  420
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cgagaagggc gggtcatacc agctgccctc catggaagct ggggtgtcga tgatgaaaag  480
tgtcggaacg tacgctagta                                              500

SEQ ID NO: 60          moltype = DNA  length = 500
FEATURE                Location/Qualifiers
source                 1..500
                       mol_type = other DNA
                       organism = Moniliella pollinis
SEQUENCE: 60
acgaagggtt tcccatctgt gtgacctgat cccttttttac gttttttttgt tttgtttttt  60
cttttgttca tgatcaactt gggagtgatt cctcttggtg atgtcgaaaa ttgaactgtt  120
ataccaatat tctgttgcct ttgtttcctg tttgtagttt actacttttt acagtagtcg  180
ttgttcctgc cttgcttgcg tttgtcaggg tttcccttta gcttagtacc aatatctagc  240
cttaattttg cttattcttt ctcaccgccc atgattgatt tcaatcatga ttaatgattt  300
atgcatgaaa actcaaaata ctcaattgtc ttgtacaagg caatcacaag ctggtcagcc  360
attctcagcg cctataagag ctttcagcca ggccggcccc aaaacgatct accctgtgga  420
cctggctagg ctaaactttg gaaagtcatg caattatgaa cgtttgaagc tcaagcgctc  480
ctaccctctc ctggagcagt                                              500

SEQ ID NO: 61          moltype = DNA  length = 500
FEATURE                Location/Qualifiers
source                 1..500
                       mol_type = other DNA
                       organism = Moniliella pollinis
SEQUENCE: 61
tggtgttcac cactgttgct tccccctttgt tggagcgtcc gtcgttctga tggcgctctg  60
agcccaaaat atcccttctt cgcccgagag caagcctcgg agtgtgcttg caatctactt  120
ttcctactaa agttatcttc ttgtggcttc tgaatttgaa tagaacgtgt tcagacttgt  180
tagctaaatt tgttgatgcg agatacttga cagaatcccg ttgatgaact catctgtgtg  240
accaccgcct ggtgatggaa tctcgcgcgg aagccgtggt gttactgact ctcaagcaat  300
tcagtagata aatttgtata acatcattgc aaggggtgct agcaatcaat tgtccaatac  360
caaacggcgg caagtgcaaa ggggcgatag agagtttctg cgaggaaggg aaggaatccg  420
tatgcctgaa aacttttgta caagtgtctt tgtaagggcg aggtctgctc tgacatctct  480
ctgagatctg agatctgata                                              500

SEQ ID NO: 62          moltype = DNA  length = 500
FEATURE                Location/Qualifiers
source                 1..500
                       mol_type = other DNA
                       organism = Moniliella pollinis
SEQUENCE: 62
gtttcttggc tgttgacgaa atcttatgaa atgcacgggt ctaatatttc ctgccaattg  60
tgtacccatc tcgtttgtga atgtcatagc tagctctgta gtttctaaac gaccgtctta  120
ggccggggtt tgctgattgg ggattaagga taggatactt ggccagtttg ggccgatcaa  180
tatcaatctg ctaggcgttc cttaggtcaa cttgaattat gatatatgag gtatgaggaa  240
agtgggctaa atctgtggat taggatcaga aaaaacgtgg cagaccctcg atctaggcct  300
acgtcagcaa tttcagcaat ttcagcgcct tgccccttag ataggtttca gccgctcagc  360
gcgcgcaatt tcagcatgga tgagatcaaa ccaaggggca gcagttggcg accctgtagt  420
tccaatcgca tgcttagtgc ataacgcgca catcaacgtg tcagattctc agagatctca  480
acaagcccac tagaccctag                                              500

SEQ ID NO: 63          moltype = DNA  length = 1679
FEATURE                Location/Qualifiers
source                 1..1679
                       mol_type = other DNA
                       organism = synthetic construct
SEQUENCE: 63
aagacctctg tcccacaaac cttgatggtc tgtgaaaccg tcaacaacat cattggtaga  60
actgtcaatc caagaaacaa aaattggtcc tgtggtggtt cttctggtgg tgaaggtgct  120
attgttggta ttagaggtgg tgttattggt gtcggtactg acattggtgg ttccattaga  180
gtcccagctg ctttcaactt tttatacggt ttgagaccat ctcacggtag attgccatat  240
gctaaaatgg ctaactctat ggaaggtcaa gaaaccgttc actccgtcgt tggtcctatc  300
actcactccg tcgaagactt gagattgttc accaaatctg tcttgggtca agaaccttgg  360
aagtacgact ctaaggtcat ccccatgcca tggagacaat ctgaatctga catcattgcc  420
tctaagatta agaatggtgg tttgaacatt ggttattaca atttcgacgg taacgtcttg  480
ccacacccac caattttacg tggtgtcgaa actaccgttg ccgctttggc caaggctggt  540
cacaccgtta ctccatggac tccatacaag catgatttct gtcatgactt gatttcccac  600
atctatgctg ctgatggttc tgccgacgtc atgagagaca tttctgcctc tggtgagcca  660
gccatcccta acattaagga cttgttgaac ccaaatatta aggctgttaa catgaacgaa  720
ttgtgggaca ctcatttaca aaagtggaac tatcaaatgg aatacttgga aaagtggcgt  780
gaagctgaag aaaaagctgg taaggaattg gacgctatta tcgctccaat tactcctacc  840
gtcgctgtca gacacgatca attcagatac tacggttacg cctccgttat tagcttattg  900
gatttcacct ctgttgtcgt cccagtcact ttcgctgata agaatattga taagaagaac  960
gaatctttta aagctgtttc cgaattggat gctttggttc aagaagaata cgacccagag  1020
gcttatcacg gtgctcctgt tgctgttcaa gttattggta gaagattgtc cgaagagaga  1080
actttggcta tcgccgaaga agtcggtaaa ttgttgggta acgtcgtcac tccataatgg  1140
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tgttcaccac tgttgcttcc cctttgttgg agcgtccgtc gttctgatgg cgctctgagc  1200
ccaaaatatc ccttcttcgc ccgagagcaa gcctcggagt gtgcttgcaa tctacttttc  1260
ctactaaagt tatcttcttg tggcttctga atttgaatag aacgtgttca gacttgttag  1320
ctaaatttgt tgatgcgaga tacttgacag aatcccgttg atgaactcat ctgtgtgacc  1380
accgcctggt gatggaatct cgcgcggaag ccgtggtgtt actgactctc aagcaattca  1440
gtagataaat ttgtataaca tcattgcaag ggtgctagc aatcaattgt ccaataccaa  1500
acggcggcaa gtgcaaaggg gcgatagaga gtttctgcga ggaagggaag gaatccgtat  1560
gcctgaaaac ttttgtacaa gtgtctttgt aagggcgagg tctgctctga catctctctg  1620
agatctgaga tctgatacct gcaggataac ttcgtataat gtatgctata cgaagttat   1679

SEQ ID NO: 64           moltype = DNA  length = 1903
FEATURE                 Location/Qualifiers
source                  1..1903
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 64
ataacttcgt ataatgtatg ctatacgaag ttatcctgca ggagctcaga ccttgcatgt  60
tagccttgac cttgcctcta tgaccactaa ttcgcacaag attgcacgtt accattgtca  120
tcttgccatg tatccaatgt ttcctgtttg ccatagtacg tagtagatac atcgatcatc  180
gatgcagatg acgcagatgg cattggttcg ctttgcccta tgcatgcatc atgcatcatg  240
catcatgcgt catgcgtcac tatccatatt aatcattgcc ctctgtacta ggccatgaca  300
cctgcagacc cttccccagg gatgcttctt gtaatagcca atatactaca caactttacc  360
gattacgatc gagatcgatc tcactatagt ttgtcgaact tcgatgcaca tcgagggtgc  420
ggaaaaaagc tgacggaagc ggaagtgggt gaattggtta atttagttag attggcaggg  480
ttgttcccaa gattcaacgc aagctgagaa agaaaacaga aacacgaggc ataaggccag  540
tcagaaaatg cattacgagg ggagttgaag tgactgtggg ccagtgcctg ggtcgtcatg  600
tgcaaatttt ttcttcggtt agactgccgc agcatgcagc aactctgccg gccctatata  660
actagtgctt catcttcttg tcattcttgt ctctttacca acacaccggt actctttcac  720
catgccacaa tcctgggaag aattggccgc cgacaaacgt gcccgtttgg ctaaaaccat  780
tcctgacgaa tggaaggttc aaactttgcc tgccgaagat tccgttattg atttcccaaa  840
gaagtccggt attttgtctg aggctgaatt gaagattacc gaagcctctg ctgctgattt  900
ggtttccaag ttggccgctg gtgagttgac ttctgttgaa gtcactttgg ctttttgtaa  960
gagagctgct attgctcaac aattaaccaa ctgtgctcac gaattcttcc cagatgctgc  1020
tttagctcaa gctagagaat tagatgaata ctacgctaag cataagcgtc cagttggtcc  1080
attacacggt ttaccaatct ctttaaagga ccaattgcgt gttaagggtt acgaaacctc  1140
catgggttac atttcctggt taaacaaata cgatgaaggt gattccgtct taaccaccat  1200
gttgagaaaa gctggtgctg ttttctacgt taagacctct gtcccacaaa ccttgatggt  1260
ctgtgaaacc gtcaacaaca tcattggtag aactgtcaat ccaagaaaca aaaattggtc  1320
ctgtggtggt tcttctggtg gtgaaggtgc tattgttggt attagaggtg gtgttattgg  1380
tgtcggtact gacattggtg gttccattag agtcccagct gctttcaact tttatacgg   1440
tttaagacca tctcacggta gattgccata tgctaaaatg gctaactcta tggaaggtca  1500
agaaaccgtt cactccgtcg ttggtcctat cactcactcc gtcgaagact tgagattgtt  1560
caccaaatct gtcttgggtc aagaaccttg gaagtacgac tctaaggtca tccccatgcc  1620
atggagacaa tctgaatctg acatcattgc ctctaagatt aagaatggtg gtttgaacat  1680
tggttattac aatttcgacg gtaacgtctt gccacaccca ccaattttac gtggtgtcga  1740
aactaccgtt gccgctttgg ccaaggctgg tcacaccgtt actccatgga ctccatacaa  1800
gcatgatttc tgtcatgact tgatttccca catctatgct gctgatggtt ctgccgacgt  1860
catgagagac atttctgcct ctggtgagcc agccatccct aac                   1903

SEQ ID NO: 65           moltype = DNA  length = 876
FEATURE                 Location/Qualifiers
source                  1..876
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 65
agatgtagcc ggcgctgtag agttttggac ggataggttg ggtttcagtc gagacttcgt  60
tgaagatgac tttgcaggcg tcgtgcgtga tgatgtgacc ctgtttatct cggccgttca  120
agatcaagtt gtccctgaca atacacttgc atgggtctgg gtacgcggat tagatgaatt  180
gtacgcggaa tggtctgaag ttgtatccac taactttcgt gacgcttcag gtcctgctat  240
gactgagatt ggtgaacaac cctgggggcg tgagttcgca ctccgagatc cagctggaaa  300
ttgtgttcat ttcgtagctg aggaacagga ttaatggtgt tcaccactgt tgcttcccct  360
ttgttggagc gtccgtcgtt ctgatggcgc tctgagccca aaatatccct tcttcgcccg  420
agagcaagcc tcggagtgtg cttgcaatct acttttccta ctaaagttat cttcttgtgg  480
cttctgaatt tgaatagaac gtgttcagac ttgttagcta aatttgttga tgcgagatac  540
ttgacagaat cccgttgatg aactcatctg tgtgaccacc gcctggtgat ggaatctcgc  600
gcggaagccg tggtgttact gactctcaag caattcagta gataaatttg tataacatca  660
ttgcaagggg tgctagcaat caattgtcca ataccaaacg gcggcaagtg caaaggggcg  720
atagagagtt tctgcgagga agggaaggaa tccgtatgcc tgaaaacttt tgtacaagtg  780
tctttgtaag ggcgaggtct gctctgacat ctctctgaga tctgagatct gatacctgca  840
ggataacttc gtataatgta tgctatacga agttat                           876

SEQ ID NO: 66           moltype = DNA  length = 1087
FEATURE                 Location/Qualifiers
source                  1..1087
                        mol_type = other DNA
                        organism = synthetic construct
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SEQUENCE: 66
gaagactgag tccgtctcag ataacttcgt ataatgtatg ctatacgaag ttatcctgca  60
ggagctcaga ccttgcatgt tagccttgac cttgcctcta tgaccactaa ttcgcacaag  120
attgcacgtt accattgtca tcttgccatg tatccaatgt ttcctgtttg ccatagtacg  180
tagtagatac atcgatcatc gatgcagatg acgcagatgg cattggttcg ctttgcccta  240
tgcatgcatc atgcatcatg catcatgcgt catgcgtcac tatccatatt aatcattgcc  300
ctctgtacta ggccatgaca cctgcagacc cttccccagg gatgcttctt gtaatagcca  360
atatactaca caactttacc gattacgatc gagatcgatc tcactatagt ttgtcgaact  420
tcgatgcaca tcgagggtgc ggaaaaaagc tgacggaagc ggaagtgggt gaattggtta  480
atttagttag attggcaggg ttgttcccaa gattcaacgc aagctgagaa agaaaacaga  540
aacacgaggc ataaggccag tcagaaaatg cattacgagg ggagttgaag tgactgtggg  600
ccagtgcctg ggtcgtcatg tgcaaatttt ttcttcggtt agactgccgc agcatgcagc  660
aactctgccg gccctatata actagtgctt catcttcttg tcattcttgt ctctttacca  720
acacaccggt actctttcac catggcaaag ttgacgagcg ccgtgccggt gctaaccgcg  780
agagatgtag ccggcgctgt agagttttgg acggataggt tgggtttcag tcgagacttc  840
gttgaagatg actttgcagg cgtcgtgcgt gatgatgtga ccctgtttat ctcggccgtt  900
caagatcaag ttgtccctga caatacactt gcatgggtct gggtacgcgg attagatgaa  960
ttgtacgcgg aatggtctga agttgtatcc actaactttc gtgacgcttc aggtcctgct  1020
atgactgaga ttggtgaaca accctggggg cgtgagttcg cactccgaga tccagctgga  1080
aattgtg                                                            1087

SEQ ID NO: 67          moltype = DNA  length = 1049
FEATURE                Location/Qualifiers
source                 1..1049
                       mol_type = other DNA
                       organism = synthetic construct
SEQUENCE: 67
attgaagccc ttgatggatc attcaccaca gacacagttt ttcgcgtgac agctactggt  60
gacggattta ctctccgtga agtgcctgta gatccgccat tgacgaaagt atttccagat  120
gatgagagtg atgacgagag cgatgatggc gaagatggtg atcctgactc gcgcactttc  180
gtcgcatacg gtgatgatgg tgacctggcc ggatttgttg ttgtgtccta ctcggggtgg  240
aaccgtagat tgaccgtcga agatatcgaa gtcgcgccag aacatcgtgg tcatggcgtg  300
ggacgagcac tgatgggtct agctacggaa ttcgctcgag agcgtggggc gggtcactta  360
tggcttgaag ttaccaatgt aaatgctcct gctattcacg cttatcgccg aatgggtttc  420
accttgtgtg gcctcgacac ggcattgtat gacggtactg cttctgacgg ggagcaagcg  480
ctatacatgt ctatgccctg cccctaatgg tgttcaccac tgttgcttcc cctttgttgg  540
agcgtccgtc gttctgatgg cgctctgagc ccaaaatatc ccttcttcgc ccgagagcaa  600
gcctcggagt gtgcttgcaa tctacttttc ctactaaagt tatcttcttg tggcttctga  660
atttgaatag aacgtgttca gacttgttag ctaaatttgt tgatgcgaga tacttgacag  720
aatcccgttg atgaactcat ctgtgtgacc accgcctggt gatggaatct cgcgcggaag  780
ccgtggtgtt actgactctc aagcaattca gtagataaat ttgtataaca tcattgcaag  840
gggtgctagc aatcaattgt ccaataccaa acggcggcaa gtgcaaaggg gcgatagaga  900
gtttctgcga ggaagggaag gaatccgtat gcctgaaaac ttttgtacaa gtgtctttgt  960
aagggcgagg tctgctctga catctctctg agatctgaga tctgatacct gcaggataac  1020
ttcgtataat gtatgctata cgaagttat                                    1049

SEQ ID NO: 68          moltype = DNA  length = 1257
FEATURE                Location/Qualifiers
source                 1..1257
                       mol_type = other DNA
                       organism = synthetic construct
SEQUENCE: 68
gaagactgag tccgtctcag ataacttcgt ataatgtatg ctatacgaag ttatcctgca  60
ggagctcaga ccttgcatgt tagccttgac cttgcctcta tgaccactaa ttcgcacaag  120
attgcacgtt accattgtca tcttgccatg tatccaatgt ttcctgtttg ccatagtacg  180
tagtagatac atcgatcatc gatgcagatg acgcagatgg cattggttcg ctttgcccta  240
tgcatgcatc atgcatcatg catcatgcgt catgcgtcac tatccatatt aatcattgcc  300
ctctgtacta ggccatgaca cctgcagacc cttccccagg gatgcttctt gtaatagcca  360
atatactaca caactttacc gattacgatc gagatcgatc tcactatagt ttgtcgaact  420
tcgatgcaca tcgagggtgc ggaaaaaagc tgacggaagc ggaagtgggt gaattggtta  480
atttagttag attggcaggg ttgttcccaa gattcaacgc aagctgagaa agaaaacaga  540
aacacgaggc ataaggccag tcagaaaatg cattacgagg ggagttgaag tgactgtggg  600
ccagtgcctg ggtcgtcatg tgcaaatttt ttcttcggtt agactgccgc agcatgcagc  660
aactctgccg gccctatata actagtgctt catcttcttg tcattcttgt ctctttacca  720
acacaccggt actctttcac catgggaacg actttagatg ataccgccta ccgttatagg  780
acatcagtac cgggcgatgc agaagcaatt gaagcccttg atggatcatt caccacagac  840
acagtttttc gcgtgacagc tactggtgac ggatttactc tccgtgaagt gcctgtagat  900
ccgccattga cgaaagtatt tccagatgat gagagtgatg acgagagcga tgatggcgaa  960
gatggtgatc ctgactcgcg cactttcgtc gcatacggtg atgatggtga cctggccgga  1020
tttgttgttg tgtcctactc ggggtggaac cgtagattga ccgtcgaaga tatcgaagtc  1080
gcgccagaac atcgtggtca tggcgtggga cgagcactga tgggtctagc tacggaattc  1140
gctcgagagc gtggggcggg tcacttatgg cttgaagtta ccaatgtaaa tgctcctgct  1200
attcacgctt atcgccgaat gggtttcacc ttgtgtggcc tcgacacggc attgtat     1257

SEQ ID NO: 69          moltype = DNA  length = 1199
FEATURE                Location/Qualifiers
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source                  1..1199
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 69
cttaaacacg gcaagggtag cgttgcaaac gacgtaaccg acgaaatggt ccgtctgaat  60
tggctaactg agttcatgcc actccccacc atcaagcact tcatccgtac acctgacgat  120
gcgtggctgt tgactacggc catccccggt aagaccgctt ttcaggtcct cgaggaatac  180
cctgactcgg gcgagaacat cgtggatgcc ttggctgtgt ttttgcgccg ccttcactct  240
attccggtct gcaactgccc tttcaacagc gaccgtgttt ttaggcttgc ccaggcacaa  300
tcacgcatga acaatggtct agtcgacgct agtgatttcg atgacgagcg caacggctgg  360
ccggtggaac aagtgtggaa agagatgcat aagctcctgc ccttttcccc agactctgtc  420
gtgacgcatg gtgatttctc gcttgacaac ctcatctttg acgagggcaa gctcattggc  480
tgcattgacg tcggccgggt tggcatcgcc gatcgctacc aagatcttgc gatcctctgg  540
aattgtctgg gcgagttctc ccctagcctc cagaagcgtc tgttccagaa atatgggatc  600
gataacccag acatgaacaa gctccagttc catttgatgc tggatgaatt tttctaatgg  660
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ctactaaagt tatcttcttg tggcttctga atttgaatag aacgtgttca gacttgttag  840
ctaaatttgt tgatgcgaga tacttgacag aatcccgttg atgaactcat ctgtgtgacc  900
accgcctggt gatggaatct cgcgcggaag ccgtggtgtt actgactctc aagcaattca  960
gtagataaat ttgtataaca tcattgcaag ggtgctagc aatcaattgt ccaataccaa  1020
acggcggcaa gtgcaaaggg gcgatagaga gtttctgcga ggaagggaag gaatccgtat  1080
gcctgaaaac ttttgtacaa gtgtctttgt aagggcgagg tctgctctga catctctctg  1140
agatctgaga tctgatacct gcaggataac ttcgtataat gtatgctata cgaagttat   1199

SEQ ID NO: 70           moltype = DNA  length = 1475
FEATURE                 Location/Qualifiers
source                  1..1475
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 70
ataacttcgt ataatgtatg ctatacgaag ttatcctgca ggagctcaga ccttgcatgt  60
tagccttgac cttgcctcta tgaccactaa ttcgcacaag attgcacgtt accattgtca  120
tcttgccatg tatccaatgt ttcctgtttg ccatagtacg tagtagatac atcgatcatc  180
gatgcagatg acgcagatgg cattggttcg ctttgcccta tgcatccatc atgcatcatg  240
catcatgcat catgcgtcac tatccatatt aatcattgcc ctctgtacta ggccatgaca  300
cctgcagacc cttccccagg gatgcttctt gtaatagcca atatactaca caactttacc  360
gattacgatc gagatcgatc tcactatagt ttgtcgaact tcgatgcaca tcgagggtgc  420
ggaaaaaagc tgacggaagc ggaagtgggt gaattggtta atttagttag attggcaggg  480
ttgttcccaa gattcaacgc aagctgagaa agaaaacaga aacacgaggc ataaggccag  540
tcagaaaatg cattacgagg ggagttgaag tgactgtggg ccagtgcctg ggtcgtcatg  600
tgcaaatttt ttcttcggtt agactgccgc agcatgcagc aactctgccg gccctatata  660
actagtgctt catcttcttg tcattcttgt ctctttacca acacaccggt actctttcac  720
catgggtaag gaaaagacac acgtatcgcg accccgactc aactccaaca tggacgcaga  780
cctgtacgga tacaaatggg ctcgagataa tgtcggacag tcgggtgcga cgatctacag  840
actctatgga aaaccggacg cccccgagct ctttcttaaa cacggcaagg gtagcgttgc  900
aaacgacgta accgacgaaa tggtccgtct gaattggcta actgagttca tgccactccc  960
caccatcaag cacttcatcc gtacacctga cgatgcgtgg ctgttgacta cggccatccc  1020
cggtaagacc gcttttcagg tcctcgagga ataccctgac tcgggcgaga acatcgtgga  1080
tgccttggct gtgtttttgc gccgccttca ctctattccg gtctgcaact gccctttcaa  1140
cagcgaccgt gtttttaggc ttgcccaggc acaatcacgc atgaacaatg gtctagtcga  1200
cgctagtgat ttcgatgacg agcgcaacgg ctggccggtg gaacaagtgt ggaaagagat  1260
gcataagctc ctgccctttt ccccagactc tgtcgtgacg catggtgatt tctcgcttga  1320
caacctcatc tttgacgagg gcaagctcat tggctgcatt gacgtcggcc gggttggcat  1380
cgccgatcgc taccaagatc ttgcgatcct ctggaattgt ctgggcgagt tctcccctag  1440
cctccagaag cgtctgttcc agaaatatgg gatcg                            1475

SEQ ID NO: 71           moltype = DNA  length = 1621
FEATURE                 Location/Qualifiers
source                  1..1621
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 71
tattctggta cgagccatcc caaaaatgga tcatgacagc tgccaagagc caggactata  60
agattgagat ctacagcagc gatgacctca agtcctggaa gctcgaaagc gcctttgcga  120
acgaaggctt tcttggctat cagtatgagt gccccggcct catcgaggtg cctaccgagc  180
aagatcctag caagagctac tgggtgatgt tcatctctat caatccggga gcacccgcgg  240
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ataatcagtc aagagtcgtc gactttggca aagactacta cgccctccag accttcttta  360
acacggatcc aacgtacggt tcggcactgg gcatcgcgtg ggcgtcgaac tgggagtata  420
gcgcctttgt ccccactaac ccttggcgct cgtcgatgtc gttagtgcgt aagttctcgc  480
tgaacaccga gtaccaggca aatccggaga cagaactaat caaccttaag gcagagccta  540
tcttgaacat ctctaacgcc ggaccctggt ctaggttcgc caccaacacc acgctcacta  600
aggccaatag ttacaatgtc gatctaagca attcaacggg cacactggaa ttcgagctcg  660
tttacgcagt caacaccacc cagacgatct cgaaatccgt ctttgcagat ttgtcgctgt  720
ggttcaaggg ccttgaggac ccggaagagt acctccgcat gggttttgaa gtctcagcgt  780
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catcgttctt cctcgaccgc gggaattcta aggtaaagtt cgtgaaggaa aacccctact  840
ttaccaatcg tatgtctgtg aacaatcagc cattcaaatc ggagaacgac ctgtcctact  900
acaaggttta cggccttctc gaccaaaaca tcctcgagtt gtactttaac gacggtgatg  960
ttgtgtcgac aaacacttat ttcatgacaa cgggaaacgc tctcggatcc gtcaatatga  1020
ctacgggtgt cgacaatctg ttttacattg acaaattcca ggtgcgagaa gtcaagtaat  1080
ggtgttcacc actgttgctt cccctttgtt ggagcgtccg tcgttctgat ggcgctctga  1140
gcccaaaata tcccttcttc gcccgagagc aagcctcgga gtgtgcttgc aatctacttt  1200
tcctactaaa gttatcttct tgtggcttct gaatttgaat agaacgtgtt cagacttgtt  1260
agctaaattt gttgatgcga gatacttgac agaatcccgt tgatgaactc atctgtgtga  1320
ccaccgcctg gtgatggaat ctcgcgcgga agccgtggtg ttactgactc tcaagcaatt  1380
cagtagataa atttgtataa catcattgca aggggtgcta gcaatcaatt gtccaatacc  1440
aaacggcggc aagtgcaaag gggcgataga gagtttctgc gaggaaggga aggaatccgt  1500
atgcctgaaa actttgtac aagtgtcttt gtaagggcga ggtctgctct gacatctctc  1560
tgagatctga gatctgatac ctgcaggata acttcgtata atgtatgcta tacgaagtta  1620
t                                                                  1621

SEQ ID NO: 72          moltype = DNA  length = 1982
FEATURE                Location/Qualifiers
source                 1..1982
                       mol_type = other DNA
                       organism = synthetic construct
SEQUENCE: 72
ataacttcgt ataatgtatg ctatacgaag ttatcctgca ggagctcaga ccttgcatgt  60
tagccttgac cttgcctcta tgaccactaa ttcgcacaag attgcacgtt accattgtca  120
tcttgccatg tatccaatgt ttcctgtttg ccatagtacg tagtagatac atcgatcatc  180
gatgcagatg acgcagatgg cattggttcg ctttgcccta tgcatgcatc atgcatcatg  240
catcatgcgt catgcgtcac tatccatatt aatcattgcc ctctgtacta ggccatgaca  300
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gattacgatc gagatcgatc tcactatagt ttgtcgaact tcgatgcaca tcgagggtgc  420
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cc                                                                 1982

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                       mol_type = other DNA
                       organism = Moniliella pollinis
SEQUENCE: 73
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aagaagggat attttgggct cagagcgcca tcagaacgac ggacgctcca acaaagggga  480
agcaacagtg gtgaacacca                                              500
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1. A genetically engineered yeast cell capable of producing arabitol, the engineered yeast cell comprising:

an exogenous polynucleotide sequence encoding an arabitol 2-dehydrogeanse (ARD2DH) enzyme comprising a sequence at least 80% identical to at least one of SEQ ID NOs:1, 2, 3, 9, or 11.

2. The yeast cell of claim **1**, wherein the yeast cell is an osmotolerant yeast cell.

3. The yeast cell of claim **1**, wherein the yeast cell is a cell of the subphylum Ustilaginomycotina.

4. The yeast cell of claim **1**, wherein the yeast cell is selected from the group consisting of *Trichosporonoides megachiliensis*, Trychosporonoides *oedocephalis*, Trychosporonoides *nigrescens, Pseudozyma tsukubaensis, Trigonopsis variabilis, Moniliella*, Ustilaginomycetes, *Trichosporon, Yarrowia lipolytica, Penicillium, Torula, Pichia, Candida, Candida magnoliae*, and *Aureobasidium*

5. The yeast cell of claim **1**, wherein the yeast cell is a yeast cell of the *Moniliella* genus.

6. A genetically engineered *Moniliella* cell capable of producing arabitol, the engineered *Moniliella* cell comprising:

an exogenous polynucleotide sequence encoding an arabitol 2-dehydrogeanse (ARD2DH) enzyme comprising a sequence at least 85% identical to at least one of SEQ ID NOs:1, 2, 3, 9, or 11.

7. The yeast cell of claim **6**, wherein the cell is a *Moniliella pollinis* cell.

8. The yeast cell of claim **1**, wherein the yeast cell is capable of producing arabitol at a titer of at least 0.2 g/L when used in a fermentation process in the presence of dextrose at 35° C. for 96 hours.

9. (canceled)

10. The yeast cell of claim **1**, wherein the exogenous polynucleotide sequence is integrated into the genome of the yeast cell at a loci selected from the ER1 locus, the ER3 locus, the PDC1 locus, the pyrF locus, the TRP3 locus, the gpdIIA locus, and the gpdIIB locus.

11. The yeast cell of claim **1**, wherein the exogenous polynucleotide sequence is operably linked to a heterologous or artificial promoter selected from the group consisting of pyruvate kinase 1 promoter (PYK1p; SEQ ID NO:49), 6-phosphogluconate dehydrogenase promoter (6PGDp; SEQ ID NO:40), glyceraldehyde-3-phosphate dehydrogenase promoter (TDH3p; SEQ ID NO:42), translational elongation factor 1 promoter (TEFp; SEQ ID NO:43), modified TEFp (SEQ ID NO:41), phosphoglucomutase 1 promoter (PGM1p; SEQ ID NO:44), 3-phosphoglycerate kinase promoter (PGK1p; SEQ ID NO:45), enolase promoter (ENO1p; SEQ ID NO:46), asparagine synthetase promoter (ASNSp; SEQ ID NO:47), 50S ribosomal protein L1 promoter (RPLAp; SEQ ID NO:48), and RPL16B (SEQ ID NO:50).

12. (canceled)

13. (canceled)

14. The yeast cell of claim **1**, wherein the ARD2DH enzyme has a sequence at least 90% identical to SEQ ID NO:2, 3, 9, and/or 11.

15. The yeast cell of claim **6**, wherein the ARD2DH enzyme has a sequence at least 90% identical to SEQ ID NO: 2, 3, 9, and/or 11.

16. A method for producing arabitol, the method comprising:

contacting a substrate comprising dextrose with the engineered yeast cell of claim **1**, wherein fermentation of the substrate by the engineered yeast produces arabitol.

17. A method for producing arabitol, the method comprising:

contacting a substrate comprising dextrose with the engineered yeast cell of claim **6**, wherein fermentation of the substrate by the engineered yeast produces arabitol.

18. (canceled)

19. The method of claim **16**, wherein the fermentation temperature is at or between 25° C. to 45° C. and the volumetric oxygen uptake rate (OUR) is between 0.5 to 40 mmol $O_2/(L\cdot h)$.

20. The method of claim **16**, wherein erythritol production is reduced relative to an equivalent fermentation run with an equivalent yeast cell lacking the exogenous polynucleotide sequence.

21. The method of claim **16**, wherein erythritol production is less than 60 g/L when the fermentation is run at 35° C. for 96 hours.

22. The method of claim **16**, wherein arabitol production is at least 0.2 g/L when the fermentation is run at 35° C. for 96 hours.

23. The method of claim **16**, wherein glycerol production is reduced relative to an equivalent fermentation run with an equivalent yeast cell lacking the exogenous polynucleotide sequence.

24. The method of claim **16**, wherein ethanol production is reduced relative to an equivalent fermentation run with an equivalent yeast cell lacking the exogenous polynucleotide sequence.

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